

Overview of Machine Learning

Objectives

- What is learning?
- Types of learning
- Supervised learning
- Measuring performance

Our Schedule (Spring 2026)

Lecture Course-2026 Spring			
	ET Time (12 hrs time difference)	China Time	
Lecture 1	6-8AM on Saturday, March 14	6-8PM on Saturday, March 14	
Lecture 2	6-8:20AM on Saturday, March 21	6-8:20PM on Saturday, March 21	
Lecture 3	6-8:20AM on Saturday, March 28	6-8:20PM on Saturday, March 28	Homework 1
Lecture 4	6-8AM on Saturday, April 4	6-8PM on Saturday, April 4	Homework 2
Lecture 5	6-8AM on Saturday, April 11	6-8PM on Saturday, April 11	
Mid-term (Specific Time TBD, on the weekend of April 18/19)			
Lecture 6	6-8AM on Saturday, April 25	6-8PM on Saturday, April 25	
Lecture 7	6-8AM on Saturday, May 2	6-8PM on Saturday, May 2	
Lecture 8	6-8AM on Saturday, May 9	6-8PM on Saturday, May 9	Homework 3
Lecture 9	6-8AM on Saturday, May 16	6-8PM on Saturday, May 16	Homework 4
Lecture 10	6-8AM on Saturday, May 23	6-8PM on Saturday, May 23	
Final Term (Specific Time TBD, on the weekend of May 30/31)			

Data Science

■ Exploratory Data Analysis

- The Data Science pipeline
- Basic tools: Jupyter notebooks, Pandas
- Visualization techniques and tools
- Web scraping

■ Machine Learning

- Supervised, Unsupervised, Reinforcement approaches
- Analytics tasks: classification, regression, prediction, recommendation, clustering, optimization
- Tools: scikit-learn, Spark ML

■ Big Data

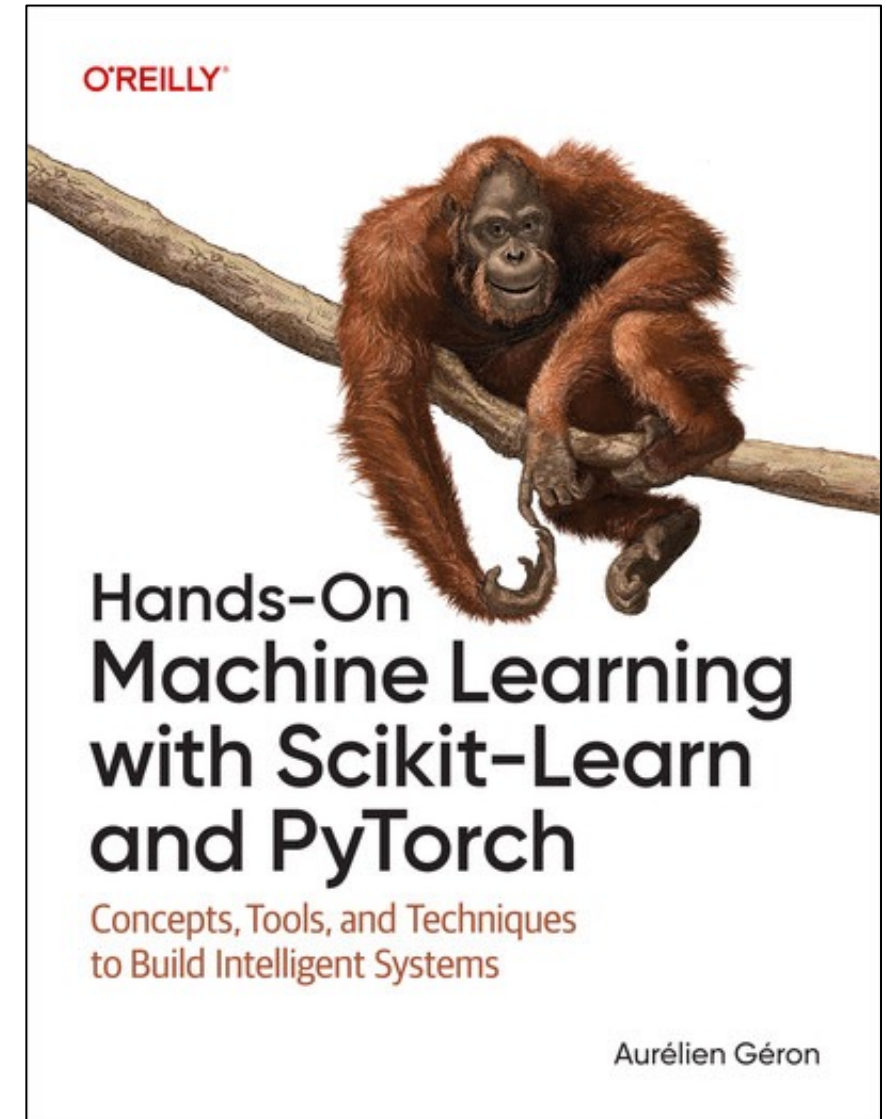
- The Hadoop platform (HDFS, map-reduce, MrJob)
- Cloud based platforms(e.g., Google Collaboratory, Azure, AWS)
- Spark (Spark SQL and Graph analytics)

Background Reading

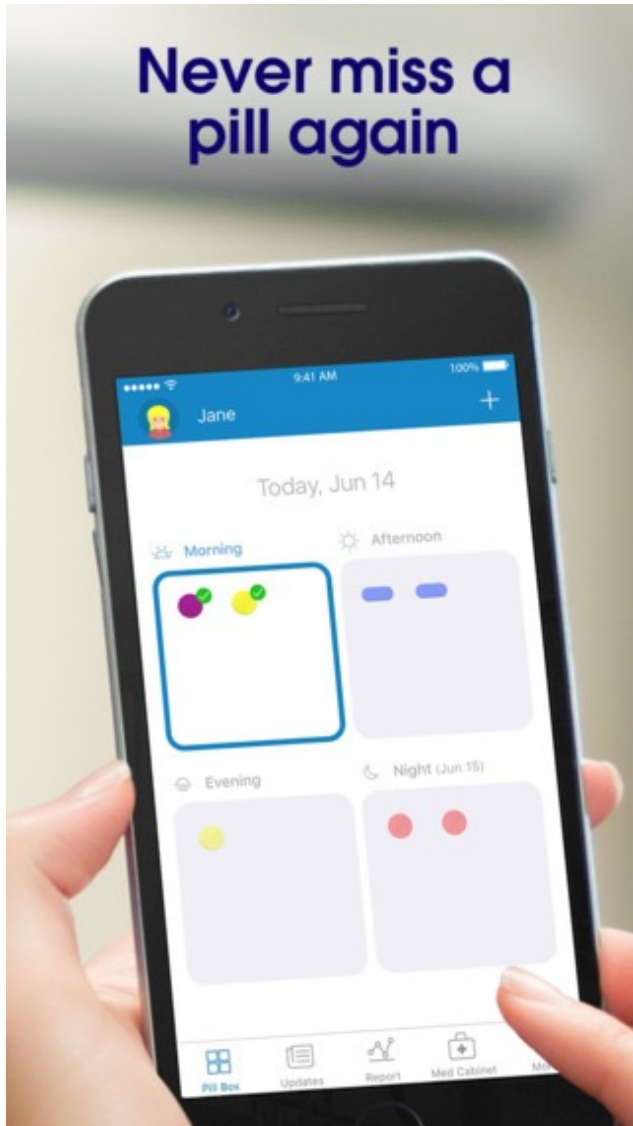
- Chapter 1: The Machine Learning Landscape
- Chapter 2: End-to-End Machine Learning Project

Machine Learning Project Checklist

1. Frame the problem and look at the big picture.
2. Get the data.
3. Explore the data to gain insights.
4. Prepare the data to better expose the underlying data patterns to machine learning algorithms.
5. Explore many different models and shortlist the best ones.
6. Fine-tune your models and combine them into a great solution.
7. Present your solution.
8. Launch, monitor, and maintain your system.

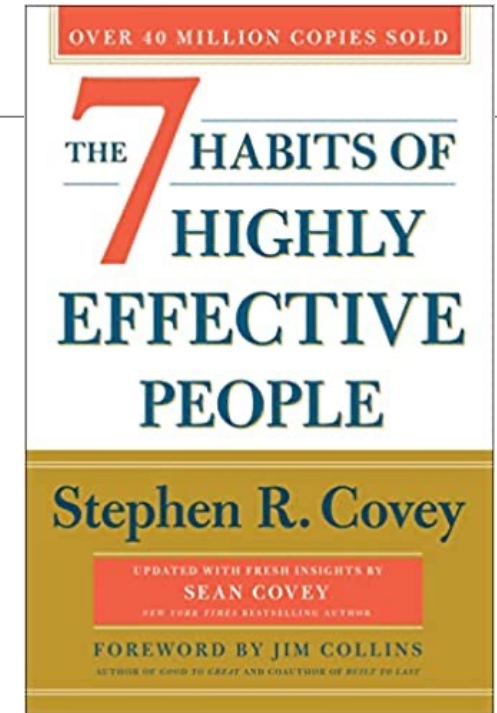
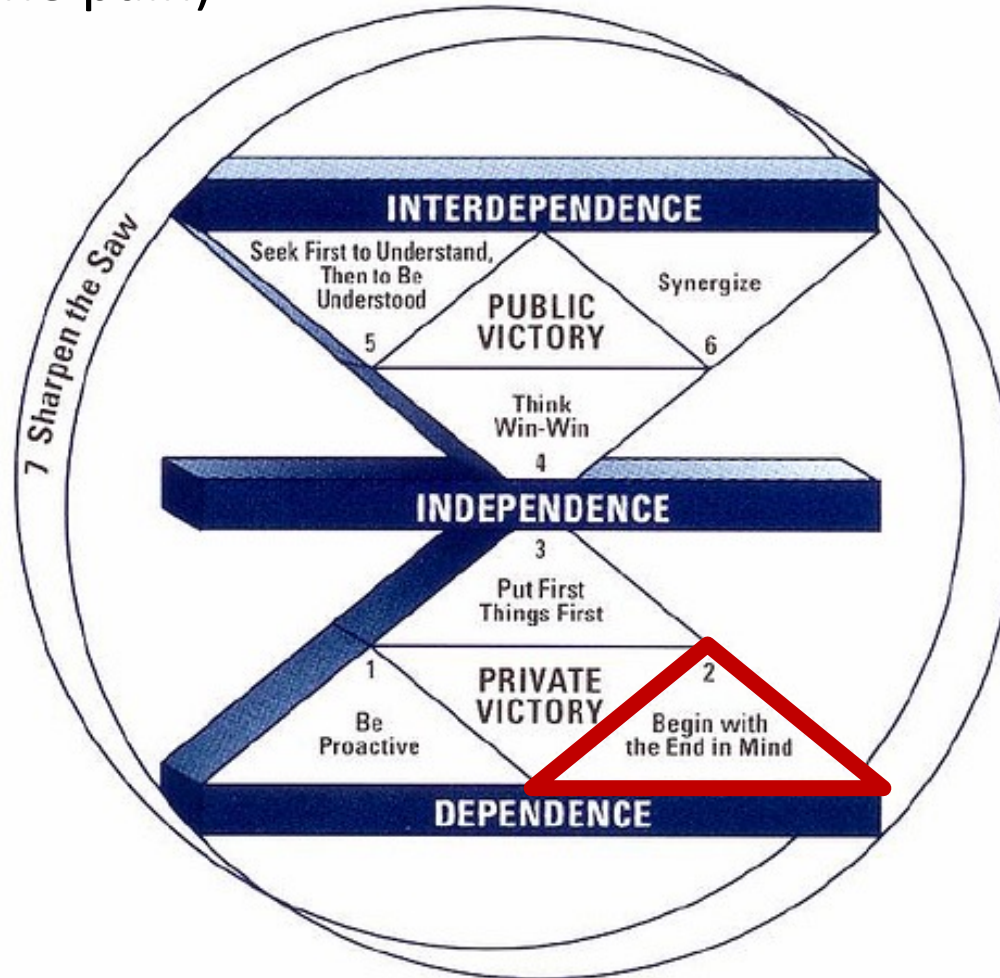


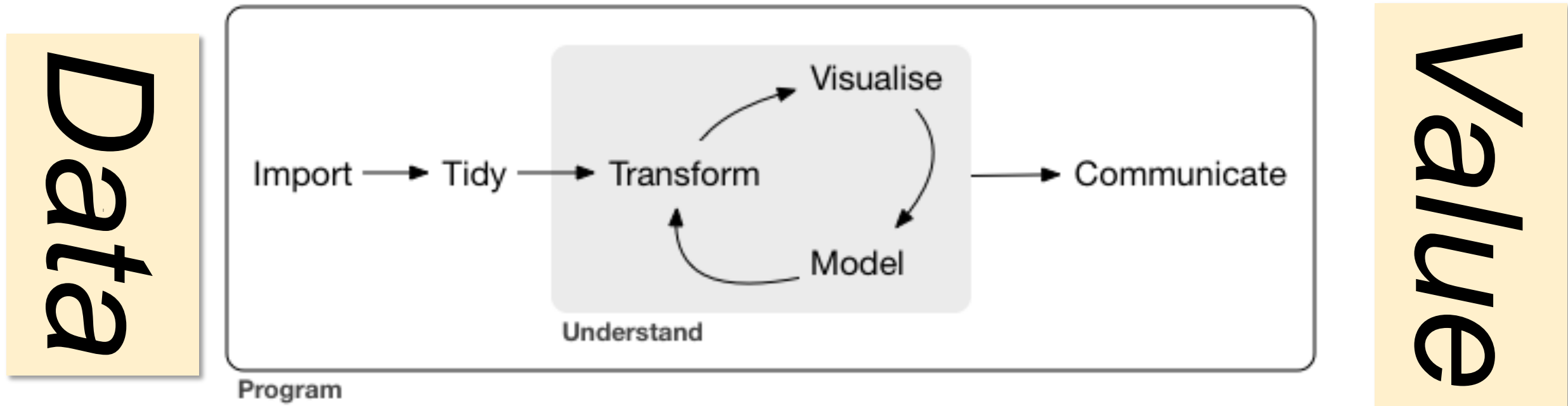
Solve the "problem" ... (it is NOT about technology)



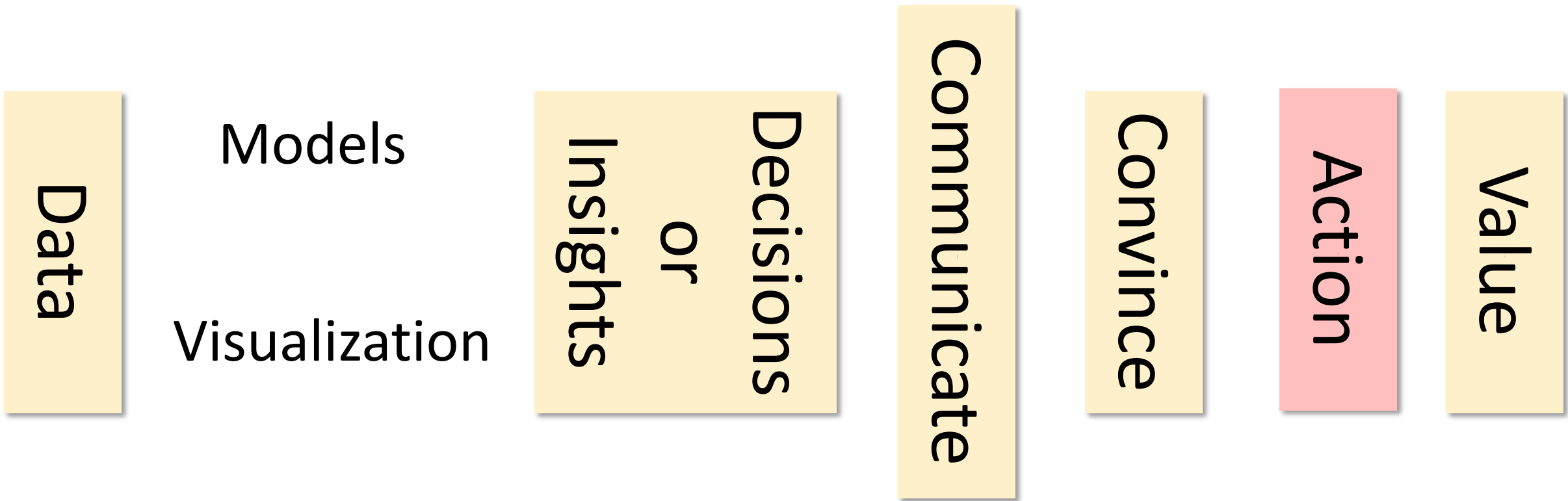
Value Proposition (end-to-end)

- Pain Point
- Problem Formulation (measure the pain)
- Solution Development
- Deployment
- Evaluation (reduction in pain)
- Maintenance / Sustainability

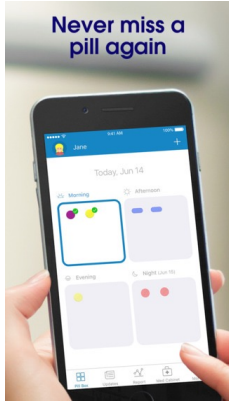




Analytics to Action



Key take aways so far

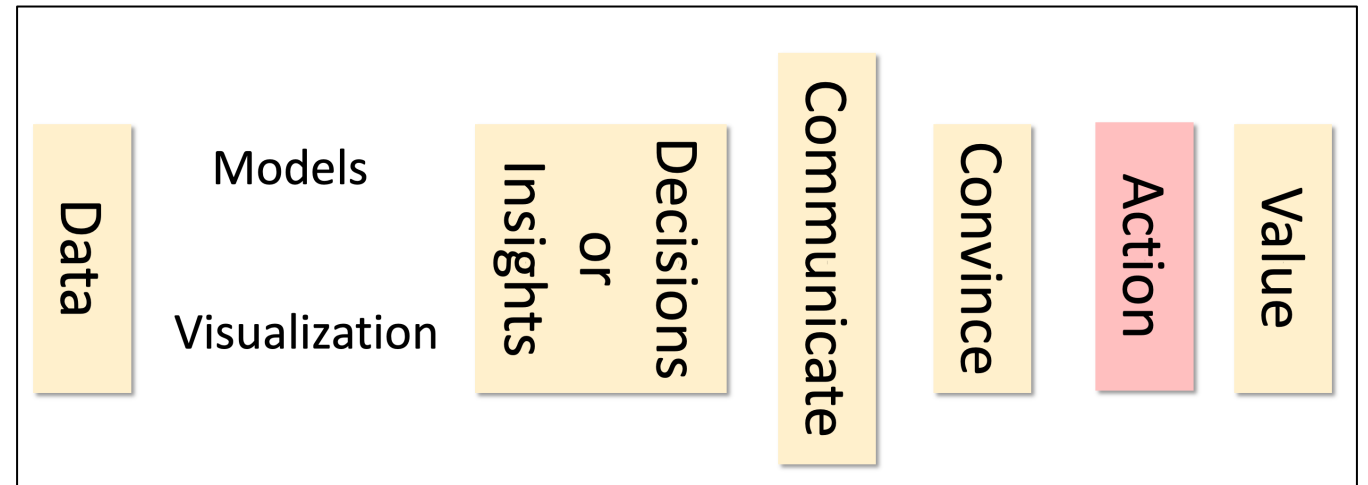


A "good" solution

- Has to be useful (solve the problem)
- Has to be useable
- Has to be economical (resource consumption - time, energy, people, ...)
- Has to be sustainable
- Should not create problems of its own

Solve the "problem" ... (it is NOT about technology)

- Pain Point
- Problem Formulation (measure the pain)
- Solution Development
- Deployment
- Evaluation (reduction in pain)
- Maintenance / Sustainability



LEARNING QUANTIFIED

What does it mean to learn?

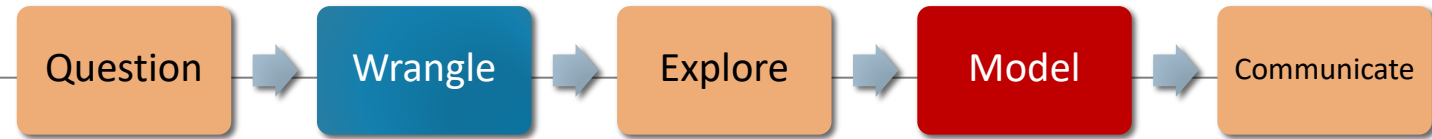
Learning Quantified

- A Computer Program is said to learn from **experience E** with respect to some **class of tasks T** and **performance measure P**, if its performance at tasks in T, as measured by P, improves with E

... Tom Mitchell

$$P(T, E+\text{more}) > P(T, E)$$

ML: Task, Technique, Tool



- **Classification / Regression**

- assign a label (classification) or numerical value (regression) to an unknown entity based on a set of features and known labels (or numerical values)

- **Clustering**

- group a bunch of entities that share common features

- **Optimization**

- from amongst a set of alternatives pick the “best” while balancing competing value metrics

- **Forecasting**

- based on the past, forecast the future

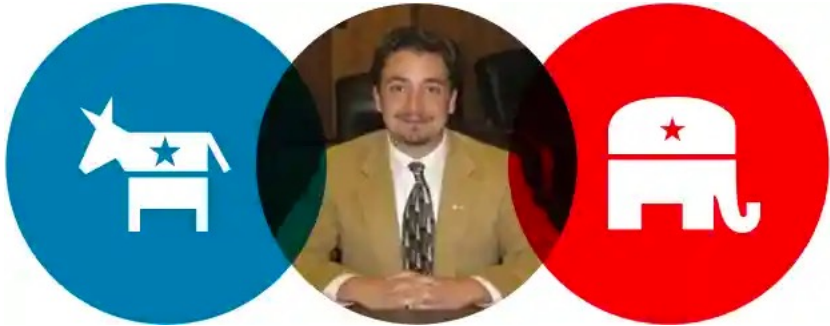
- **Recommendation**

- based on prior behavior rank order candidate preferences

- **Association**

- identify which items co-occur e.g., bread and peanut butter

AI can tell Republicans from Democrats – but can you?



Google:

*Guardian Democrat
Republican quiz*



You probably formed some rules along the lines of

- If they look young ...
- If it is woman ...
- If they are smiling / look cheerful ...
- If there is a flag in the background ...

Types of Learning

■ Supervised

- We know both the inputs and the expected outputs
- There is a ‘teacher’

■ Unsupervised

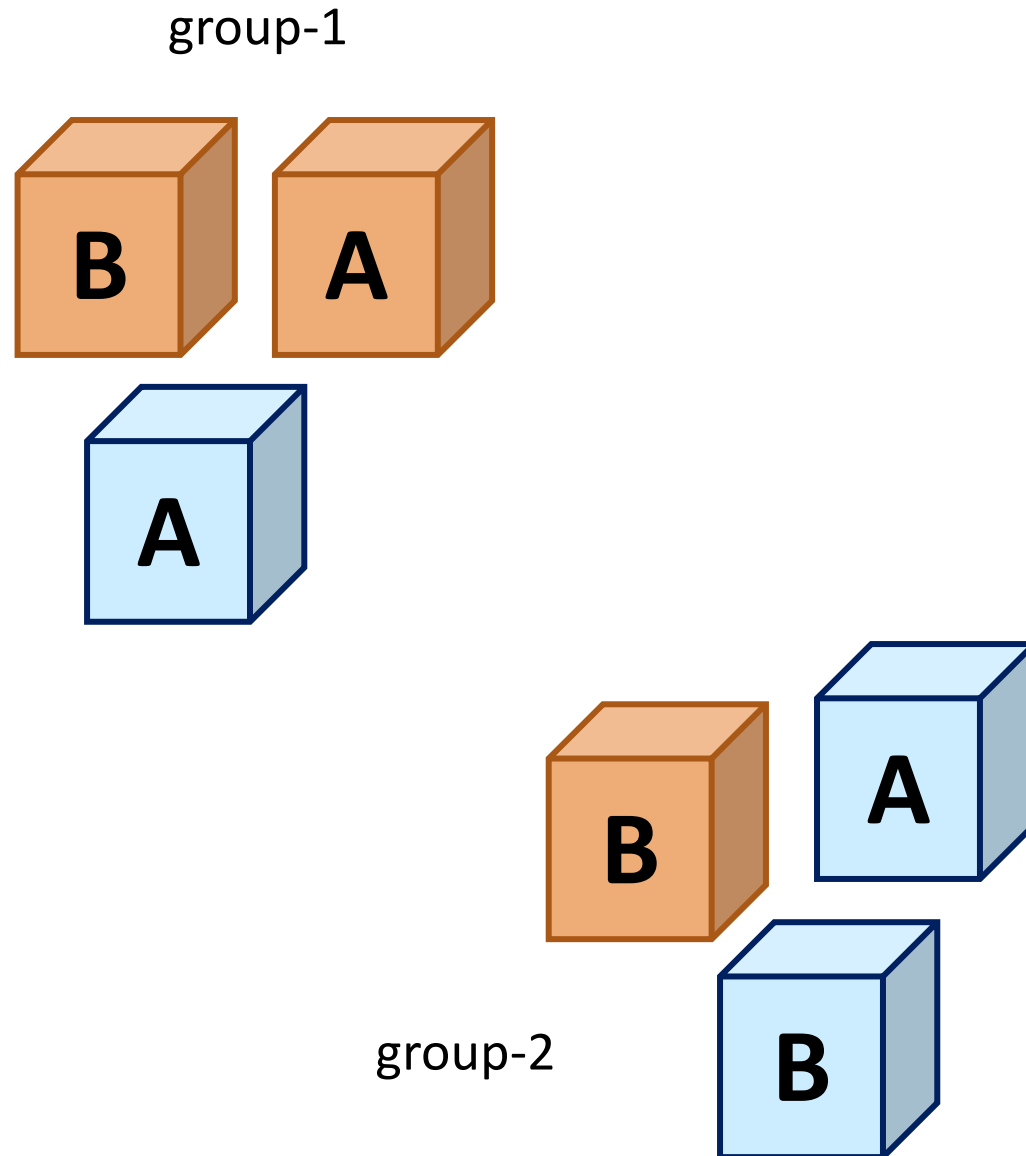
- We only have inputs

■ Reinforcement

- We know what is desired (correct) and what is not desired
- the ‘credit/blame assignment’ problem

SUPERVISED LEARNING

Supervised Learning



Two Phases

1. If during the **teaching/training phase** we say that the below block belongs to group-1



2. Then during the **testing/evaluation phase** to which group would the system assign the below block?



A SAMPLE MACHINE LEARNING TASK

Classification

The Question: Do I play tennis on a day like the below?

Outlook=*Sunny*, Temperature=*Cool*,
Humidity=*High*, Wind=*Strong*

The Question: Do I play tennis on a day like the below?

Outlook=*Sunny*, Temperature=*Cool*,
Humidity=*High*, Wind=*Strong*

*Those who can not remember the past
are condemned to repeat it.*

... George Santayana

*Those who can remember the past are
redeemed by reusing it.*

... Raja S.

Tennis / Weather Data Set

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
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Tennis / Weather Data Set

PlayTennis: training examples

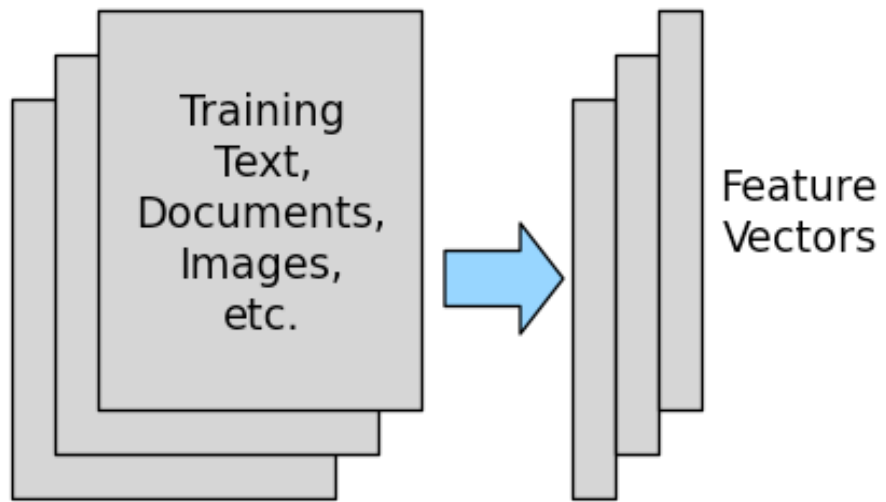
Features
X_train
y_train

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
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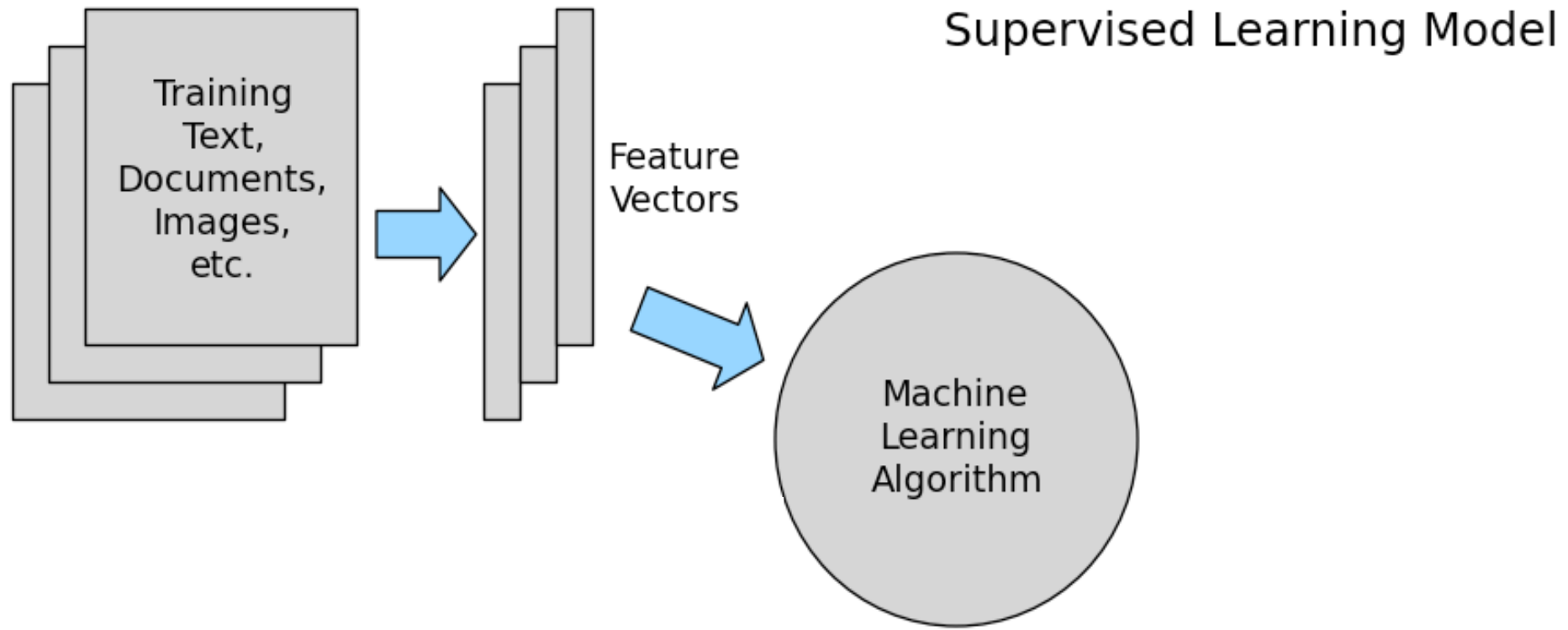
X_test: Outlook=Sunny, Temperature=Cool, Humidity=High, Wind=Strong)



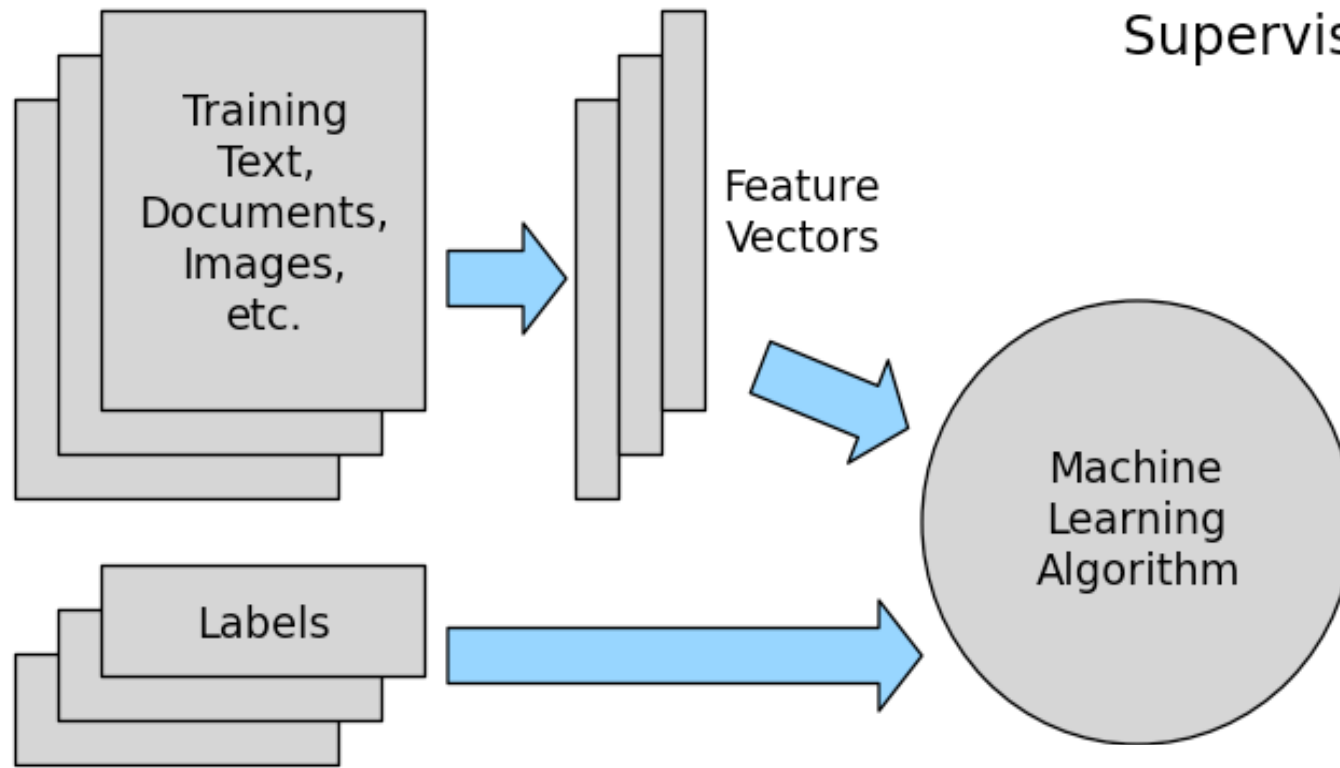
Supervised Learning Model



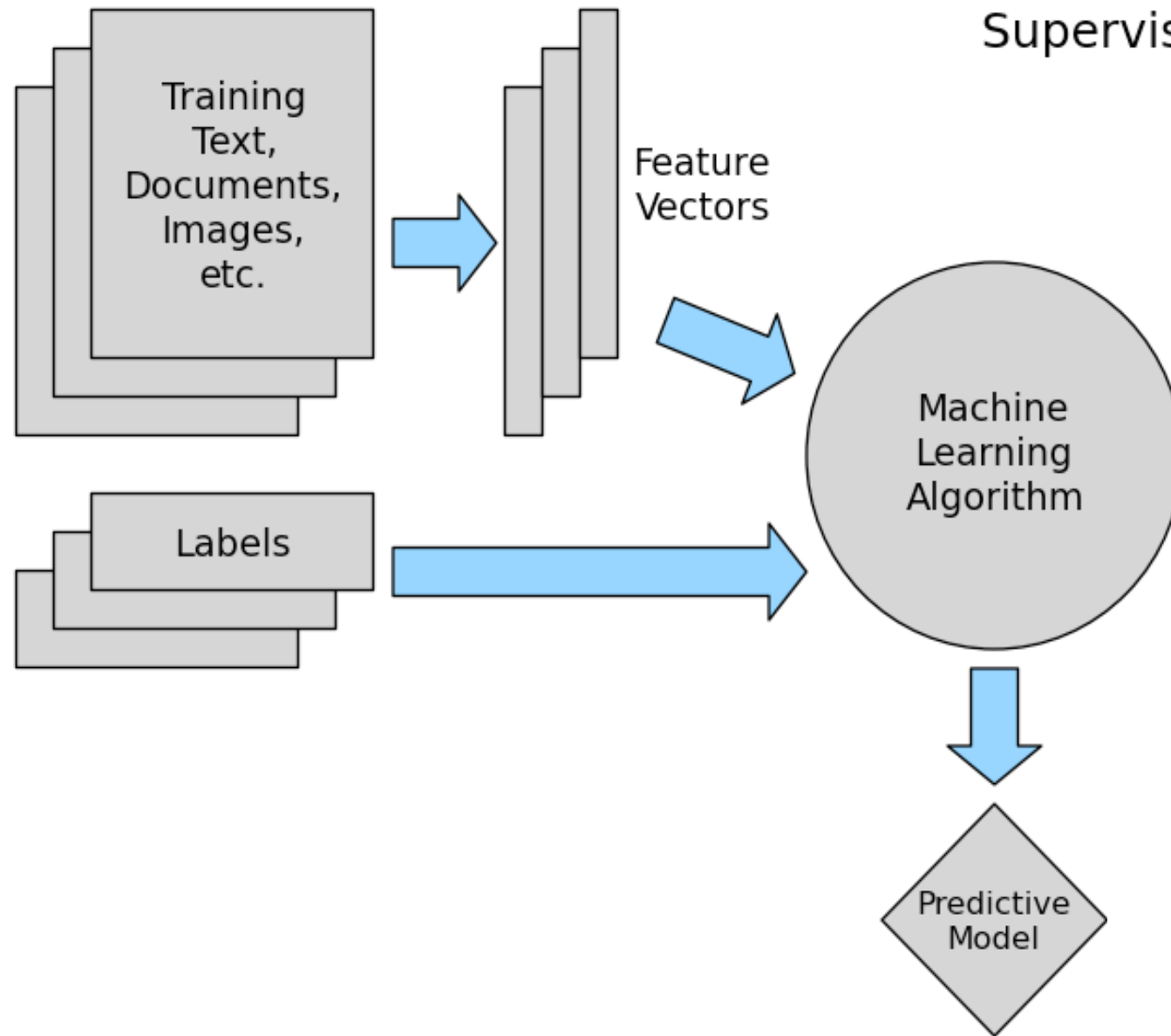
Supervised Learning Model



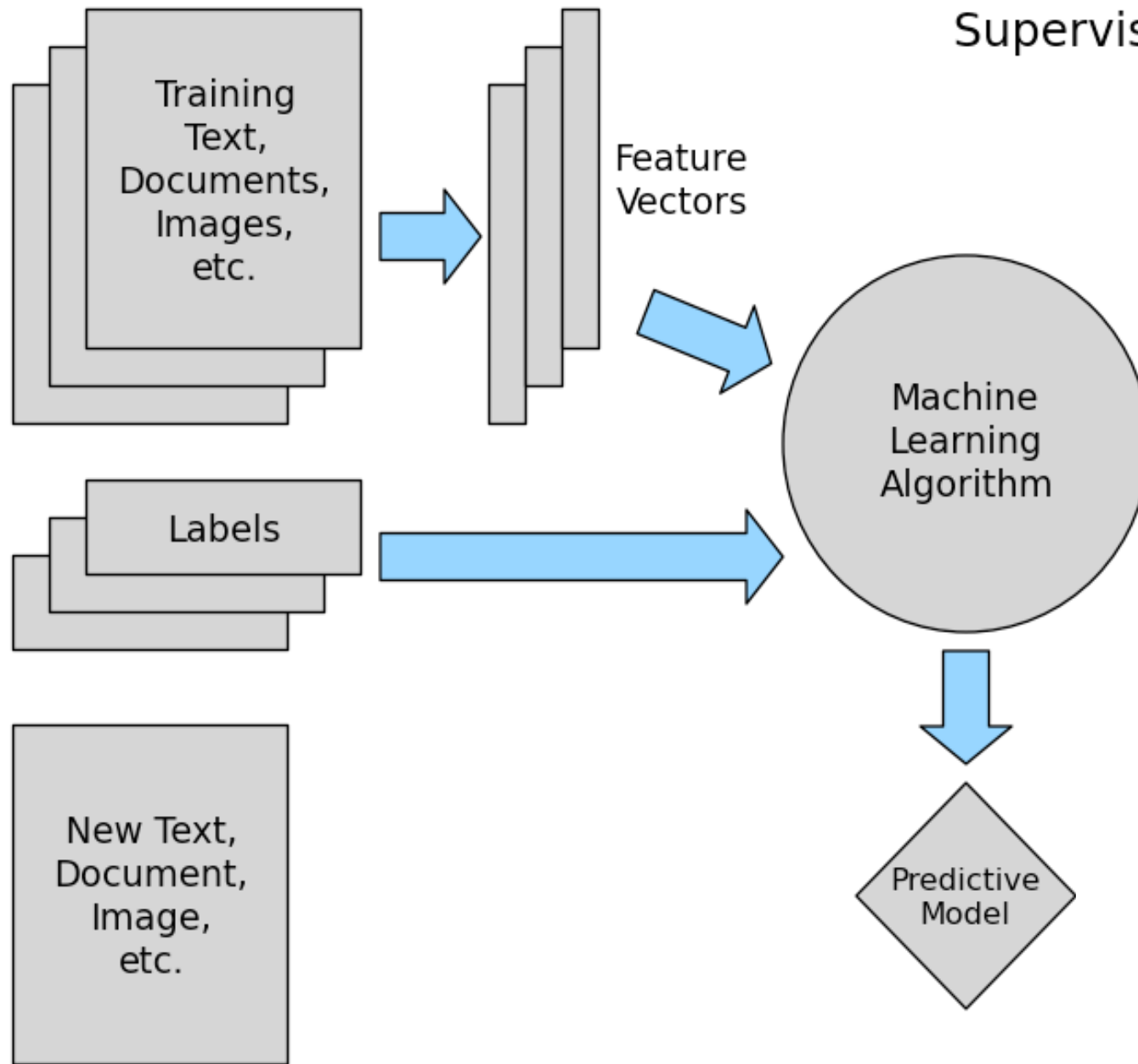
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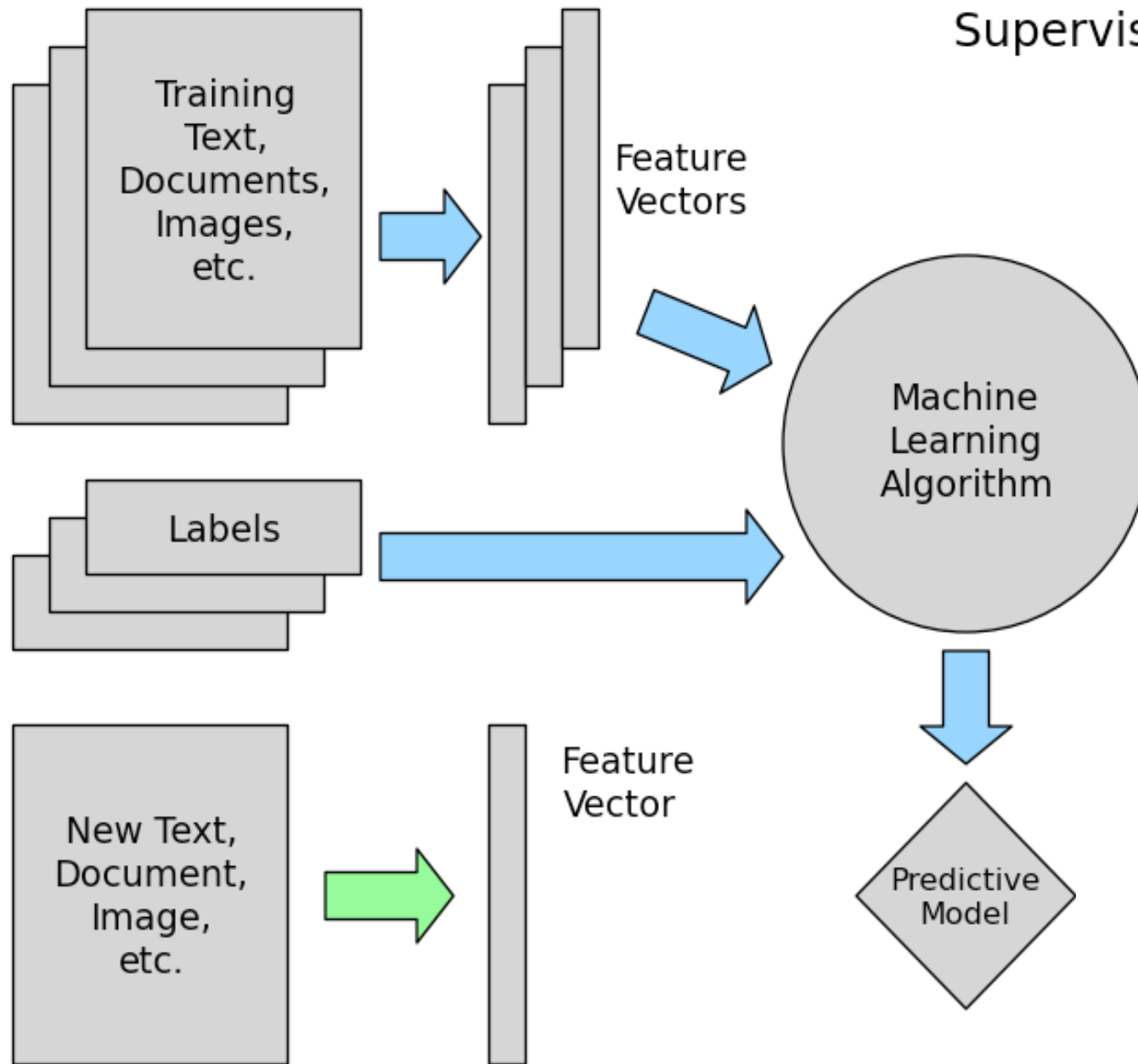
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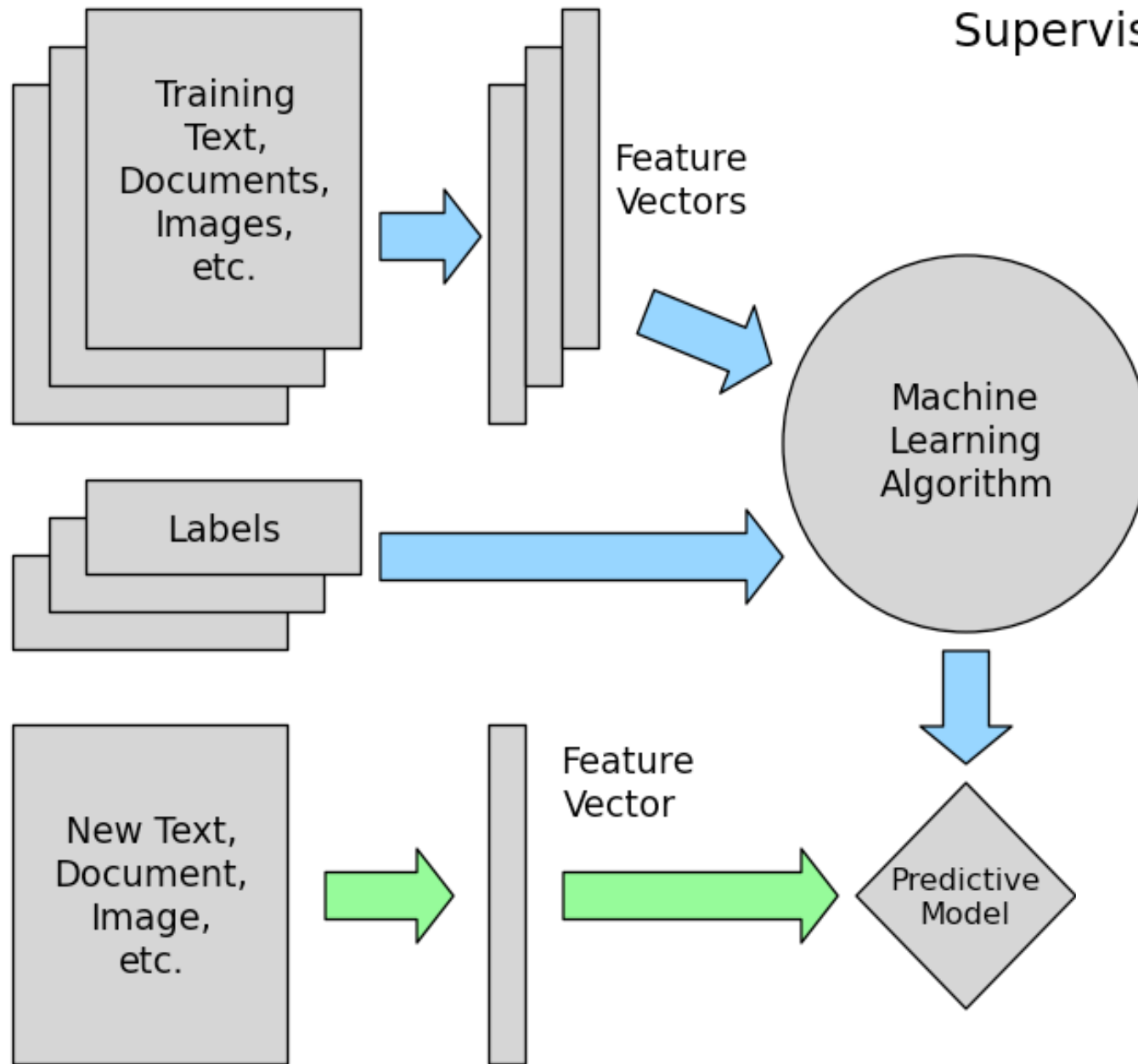
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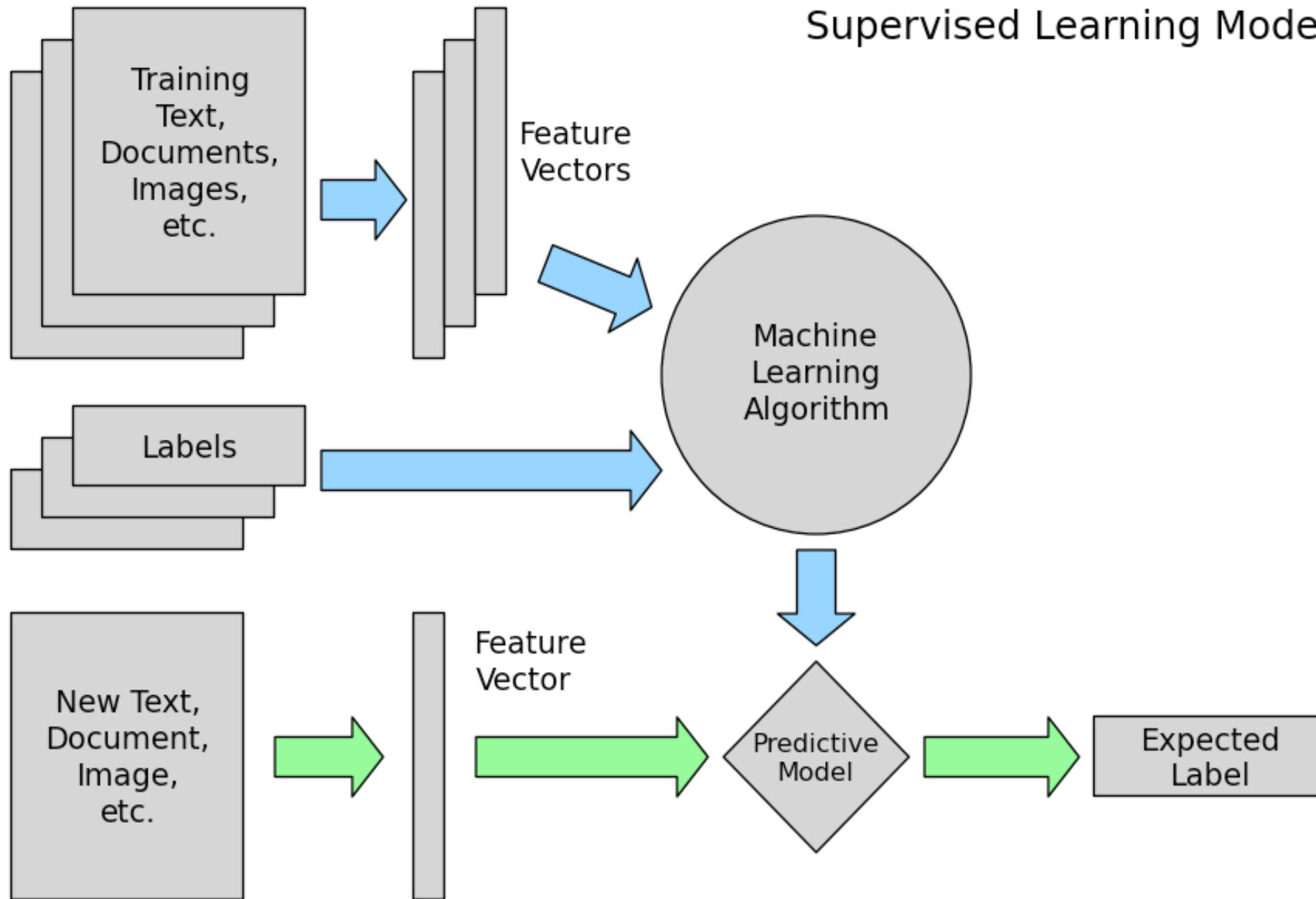
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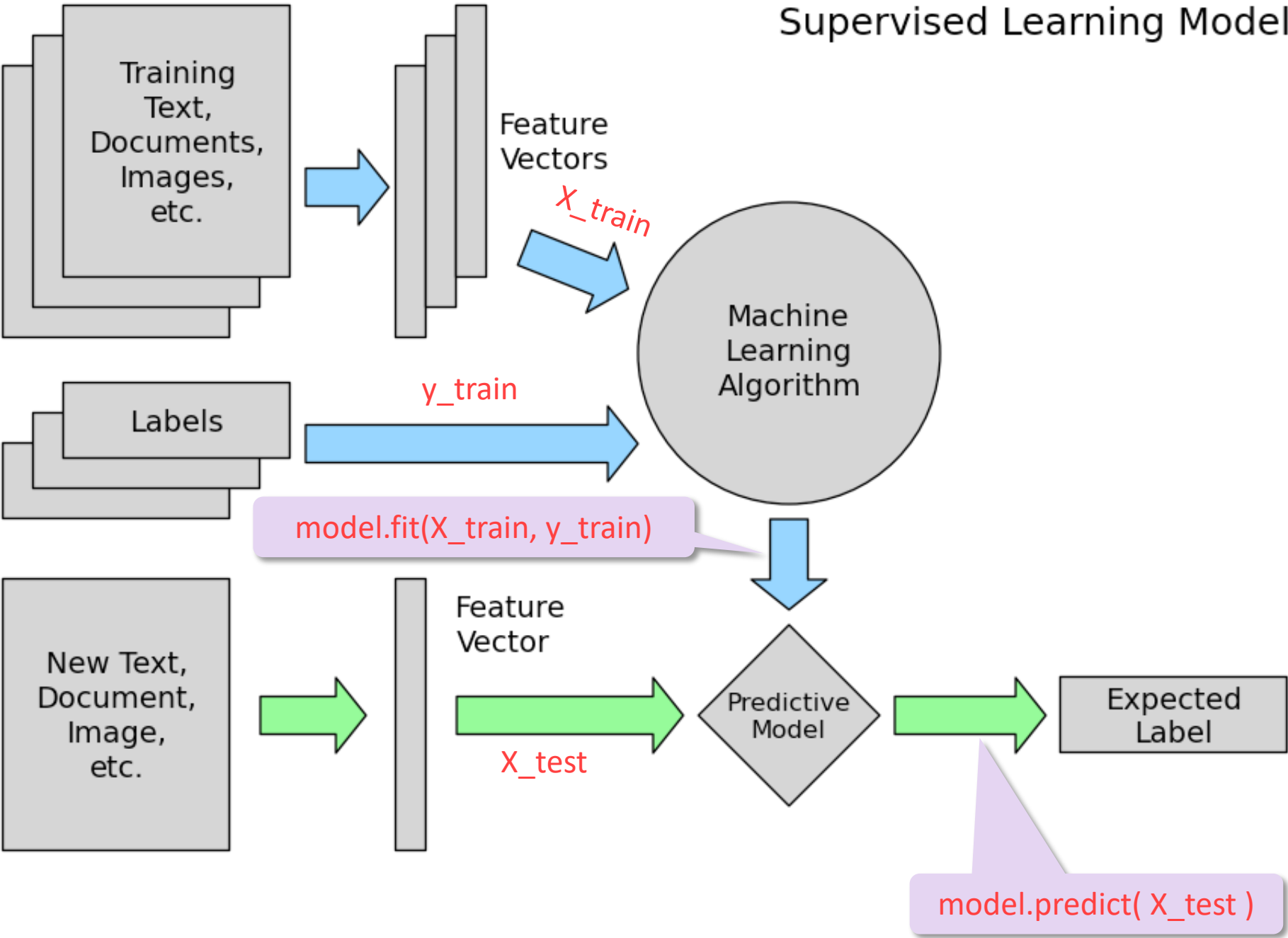
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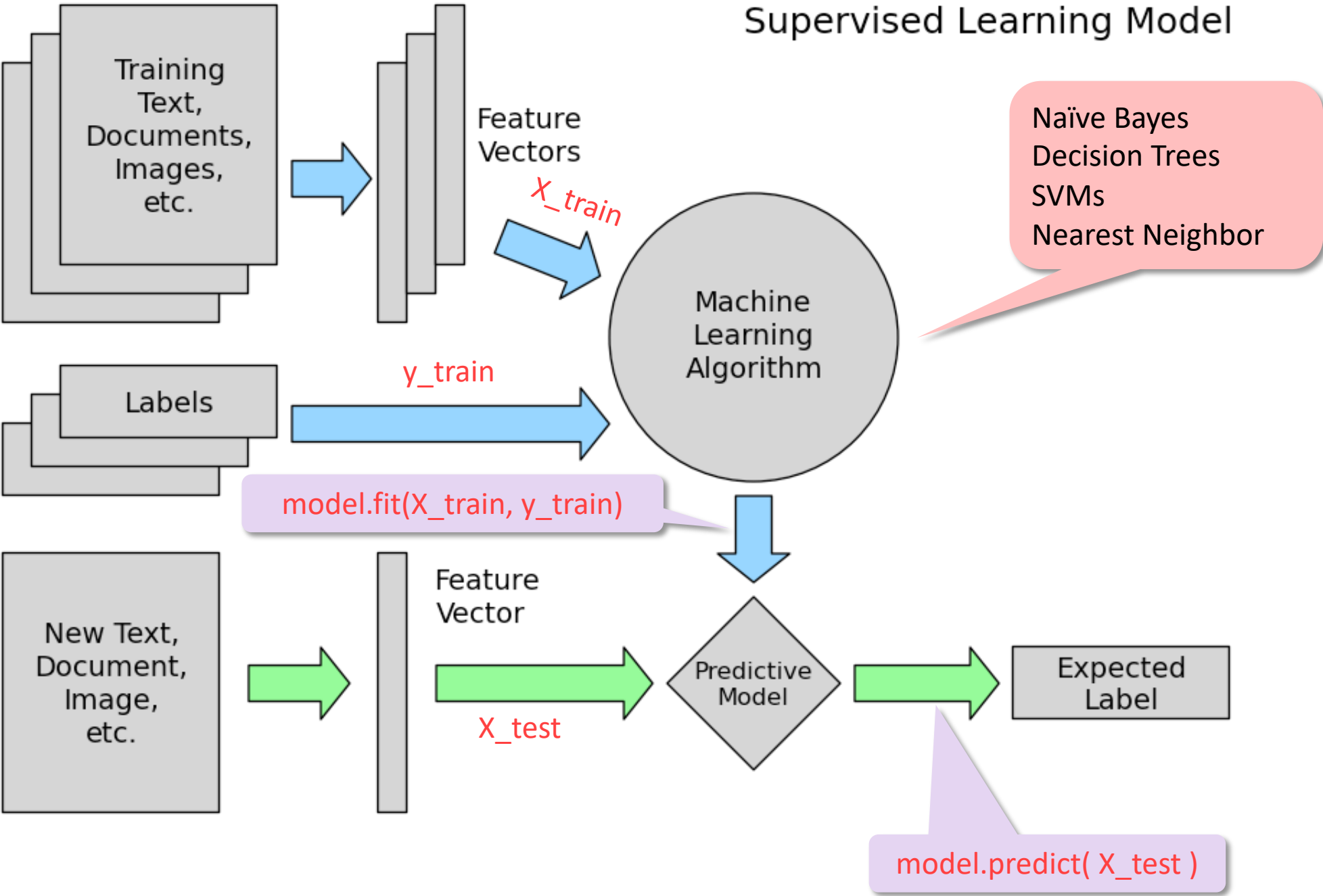
Supervised Learning Model



Supervised Learning Model



Supervised Learning Model



Bayes Rule and Learning

(The mathematics of changing your mind)

When the facts change,
I change my opinion.
What do you do, sir?
—John Maynard Keynes

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Objectives

- Motivating Bayes
- What is learning?
- Review of basic probability
- Bayes rule
- How a NB classifier works
- Bayes in the real world

If you want to build a ship, ...
... teach them to yearn for the vast and endless sea.
— Antoine de Saint-Exupéry

Objectives

Motivating Bayes

What is learning?

Review of basic probability

Bayes rule

How a NB classifier works

Bayes in the real world

MOTIVATING BAYES

3 illustrations

If Bayes' Rule is the answer

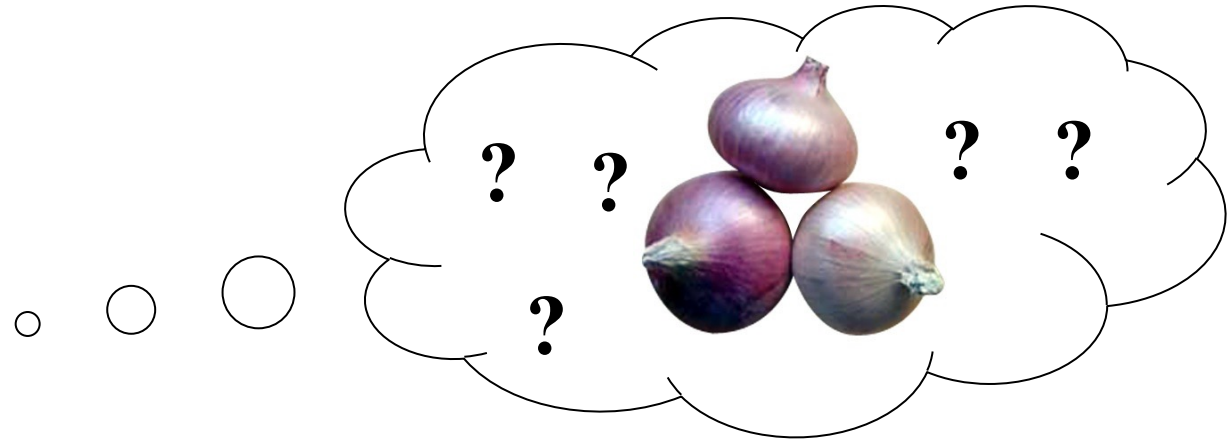
what is the question ???

Something to always consider ... when considering something new

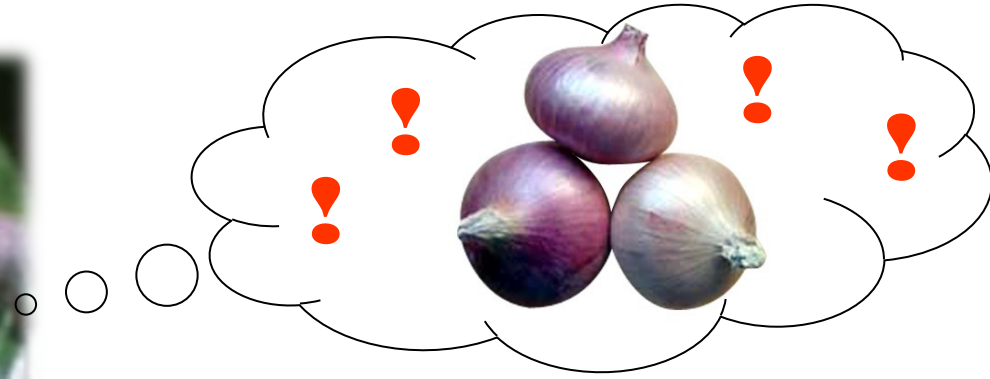
- Can we solve **new** problems?
- Can we solve **old** problems in a **better** way?

5G example

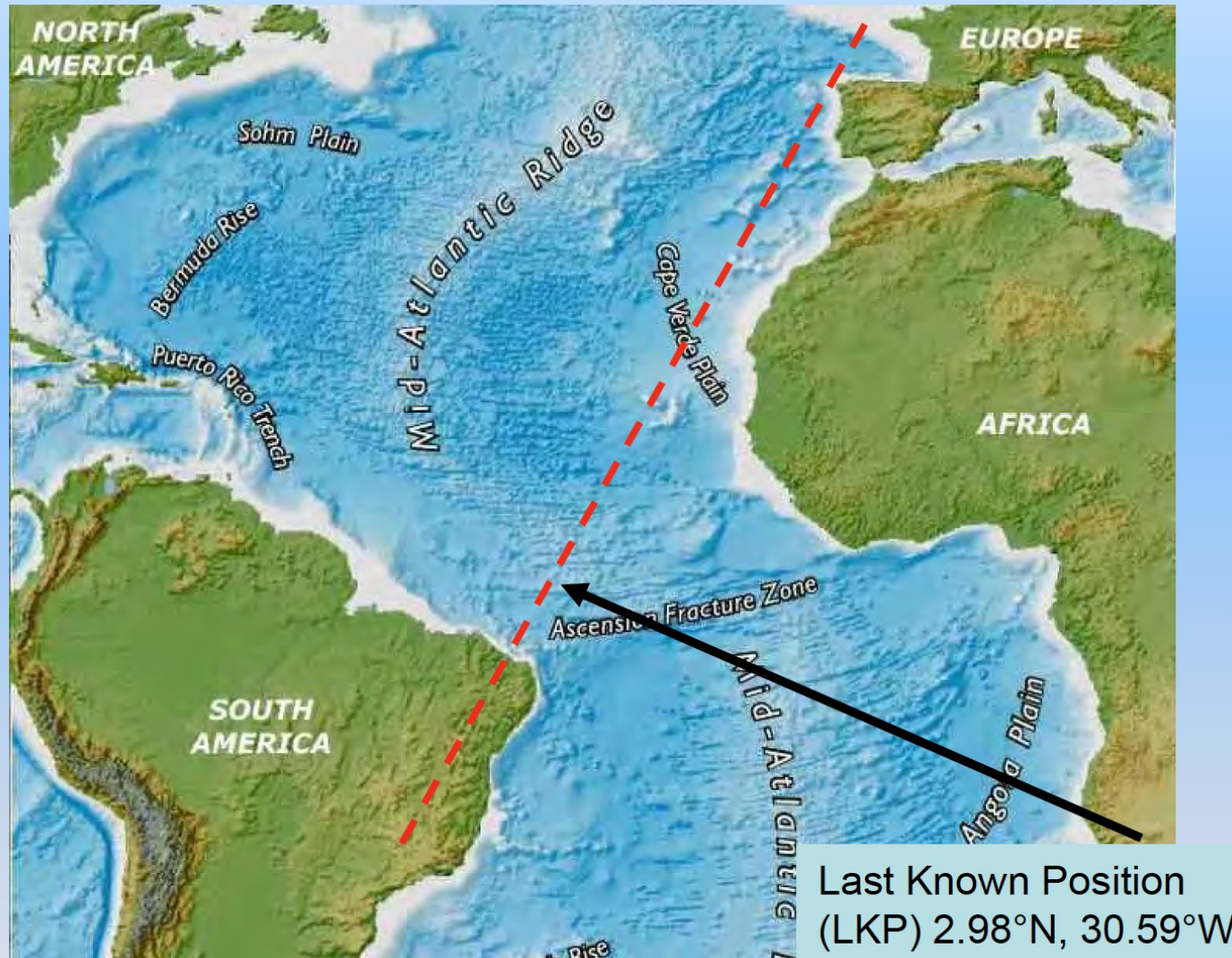
Do Gorillas like Onions?



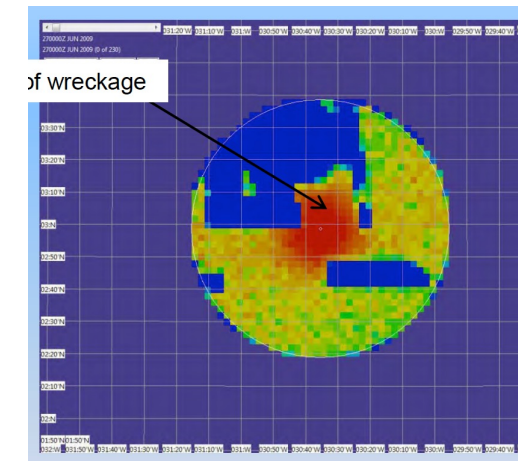
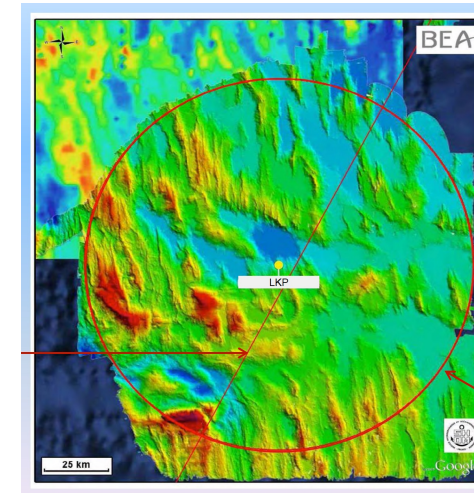
Suppose Chimps love Onions!



Location of Air France 447



- In the early morning hours of June 1, 2009, Air France Flight AF 447 from Rio to Paris with 228 passengers and crew aboard, disappeared during stormy weather over the South Atlantic
- The French Bureau of Enquiries and Analyses (BEA) after more than a year of unsuccessful search hired a company specializing in Bayesian Search.
- On April 3, 2011, an autonomous underwater vehicle (AUV) found the wreckage on the ocean bottom at a depth of 13,060 ft



Suppose you have 3 coins ...



... Two fair coins and one double headed



... Two fair coins and one double headed



What is the probability of randomly picking the double headed coin?

... Two fair coins and one double headed



You randomly pick one coin, toss it 3 times and get **3 heads**.
What is the probability now that you had picked the double headed coin?

Objectives

Motivating Bayes

What is learning?

Review of basic probability

Bayes rule

How a NB classifier works

Bayes in the Real World

LEARNING QUANTIFIED

What does it mean to learn?

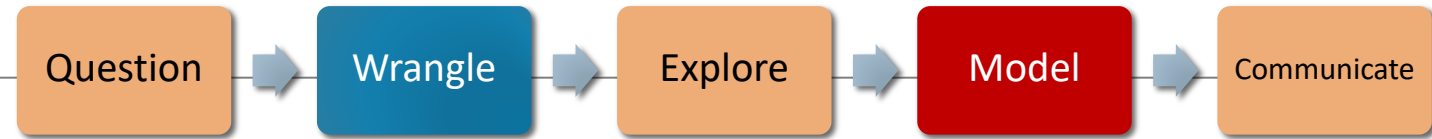
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A SAMPLE MACHINE LEARNING TASK

Supervised Learning: Classification

Tennis / Weather Data Set

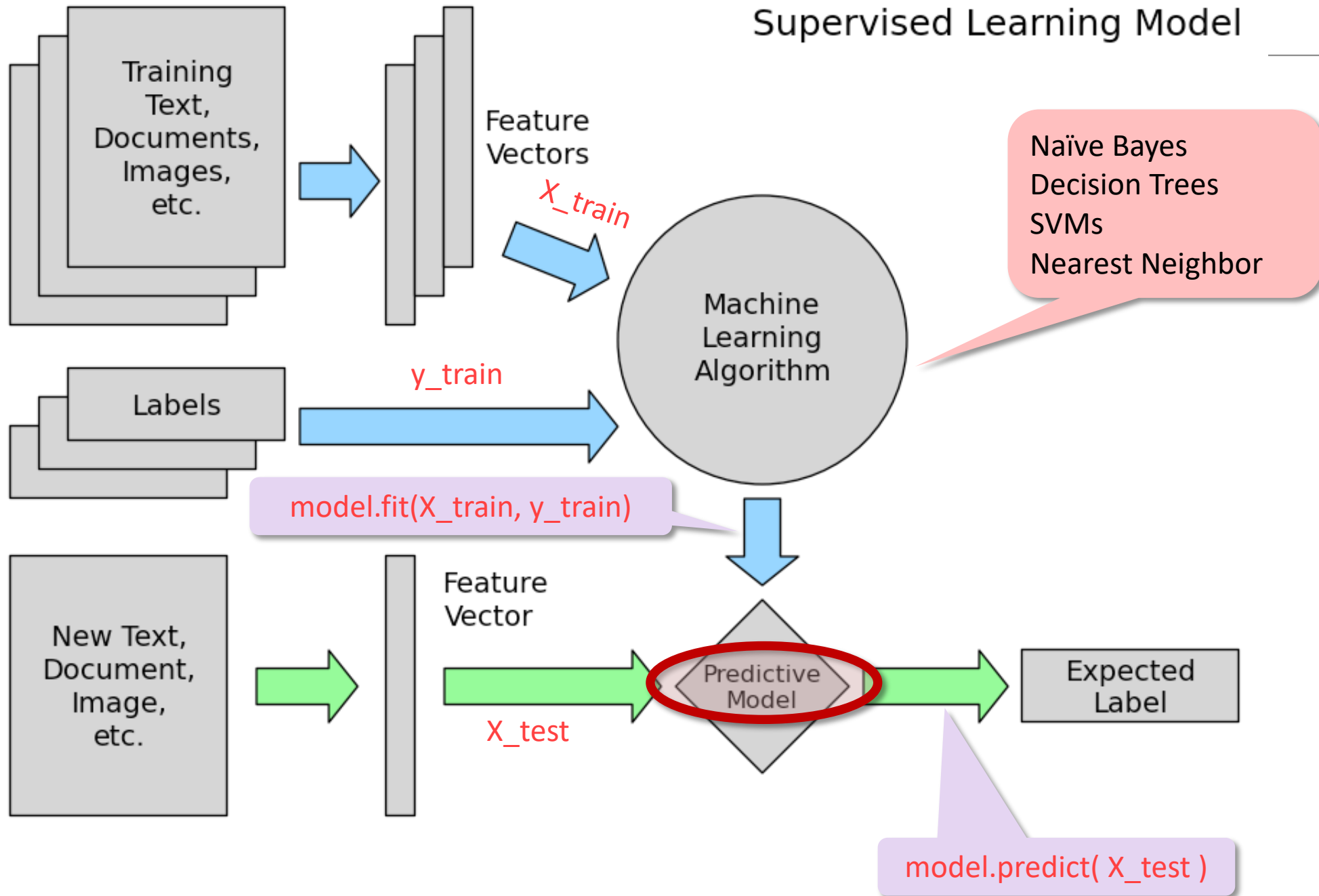
PlayTennis: training examples

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y_train

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X_test: Outlook=Sunny, Temperature=Cool, Humidity=High, Wind=Strong)

Supervised Learning Model



PROBABILITY

Terminology: Experiment, Observation, Sample Space, Event

Exercises

Objectives

Motivating Bayes

What is learning?

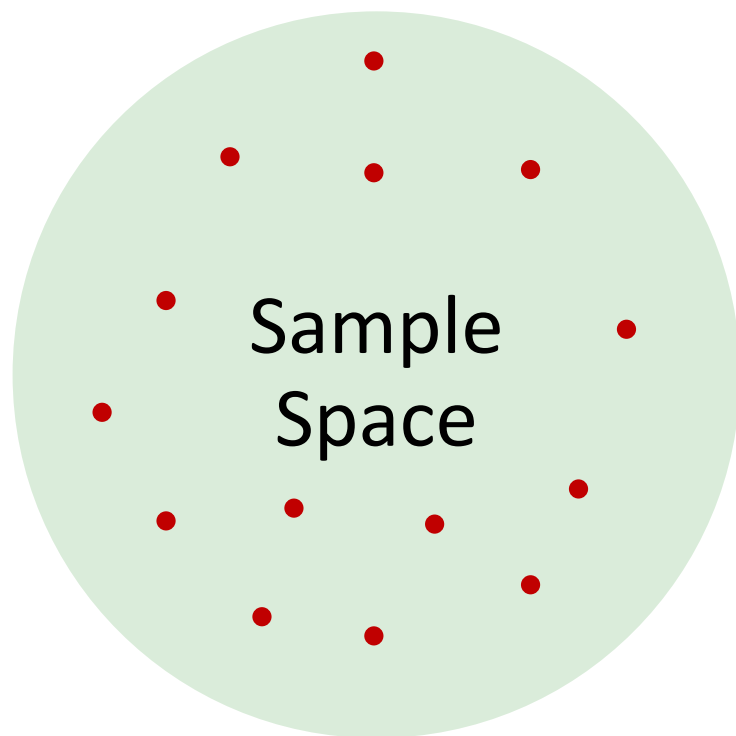
Review of basic probability

Bayes rule

How a NB classifier works

Bayes in the Real World

Sample Space: Set of all possible outcomes



Each green dot corresponds to an outcome

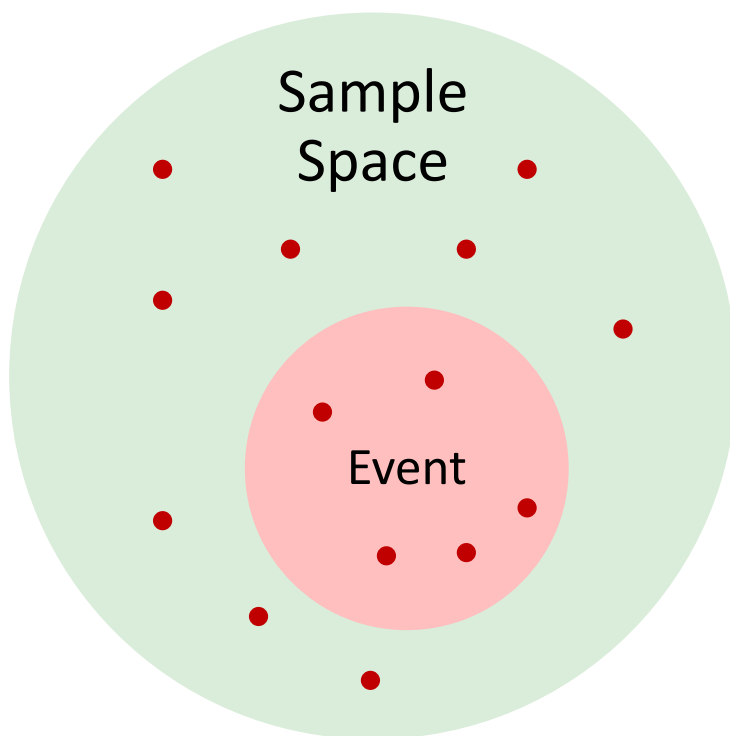
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Synonyms:

- outcome
- observation
- instance

An Event is a subset of the Sample Space



PlayTennis: training examples

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Event: (Outlook=Sunny)
{ D1, D2, D8, D9, D11 }

Classical Probability

$$\text{Probability of event E} = \frac{\text{Number of outcomes in an event E}}{\text{Total number of outcomes in Sample Space}}$$

Outcome: result of an experiment

Event: set of outcomes (subset of sample space)

Assumption: all outcomes are equally likely

$$P(\text{sunny}) = 5/14$$

Exercise

67-364 / 95-885 Data Science, Spring.

Revisiting a Historical Problem

17th century mathematicians Blaise Pascal, Pierre de Fermat, and Antoine Gombaud explored many problems in gambling, chance, and uncertainty. One of the problems that contributed towards the birth of the field of probability is the following, for which we wrote a simulation as part of assignment 1:

Which is more likely?

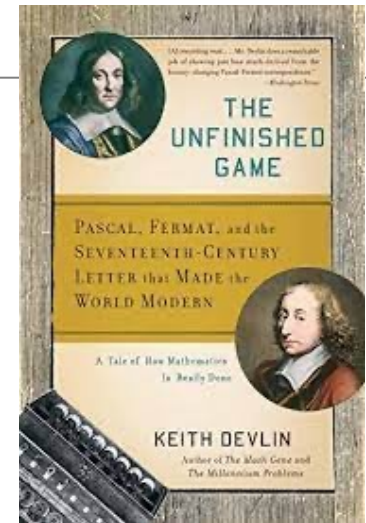
- a. a six on 4 throws of a dice
- b. two sixes on 24 throws of a pair of dice

We will solve this problem using probability with three approaches:

1. Solve by calculating the probability of the opposite event. *Calculate the probability of there not being a 6 in any throw.*
2. Solve by summing probabilities of mutually exclusive events. *The four mutually exclusive events are a six in the 1st throw, a six in the 2nd throw, a six in the 3rd throw, a six in the 4th throw.*
3. Solve by counting the number of outcomes in the event and sample space.

Counting possible outcomes is a little tricky. Here is a hint on how to go about it. When determining probability by counting we need to determine the total number of possible outcomes (the denominator) and the total number of outcomes of interest (in this case outcomes with a 6). While the experiment is to roll a single dice at most 4 times and see if you get a 6 an equivalent experiment is to simultaneously roll 4 dice and see if at least one six is rolled. If I were to simultaneously throw 4 dice, how many possible outcomes are there? This will give you the denominator. Now, how do I count the number of possible ways I can get a six? (The numerator.). A candidate roll could be 4516, another candidate roll would be 2366 (each number represents the roll of a single die). Essentially we are asking how many ways can I write the number 1—6 four times with at least 1 six? Following is a strong hint. Let the pattern 6***

- iii. Show your work and line of reasoning. You may have to use a calculator (or the Python REPL) to do the calculation.



the second die is a 6; and I don't care about the 3rd or 4th die. The roll of the presenting anything but a 6. Why don't we use the pattern *6** ? If we did counting rolls where the first die is a 6. To avoid double counting we use the v many throws fall into the pattern ++6* and finally +++6? We basically have t from here ...

et in all 3 ways will be the same and will match the result you get with the Ensure your response conforms to the following expectations:

ne problem is the key learning objective of this exercise. Make sure you solve y. calculate the probability for just "a six on 4 throws of a dice". The structure of be the same for the second experiment. You do not need to show the

Previously

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
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X '(Outlook=*Sunny*, Temperature=*Cool*, Humidity=*High*, Wind=*Strong*) = ?

Supervised Learning: Phase 1 - Training

Outlook	Temperature	Humidity	Wind
Sunny	Hot	High	Weak
Sunny	Hot	High	Strong
Overcast	Hot	High	Weak
Rain	Mild	High	Weak
Rain	Cool	Normal	Weak
Rain	Cool	Normal	Strong
Overcast	Cool	Normal	Strong
Sunny	Mild	High	Weak
Sunny	Cool	Normal	Weak
Rain	Mild	Normal	Weak
Sunny	Mild	Normal	Strong
Overcast	Mild	High	Strong
Overcast	Hot	Normal	Weak
Rain	Mild	High	Strong

Feature Matrix
X_train



get
features



ML alg.

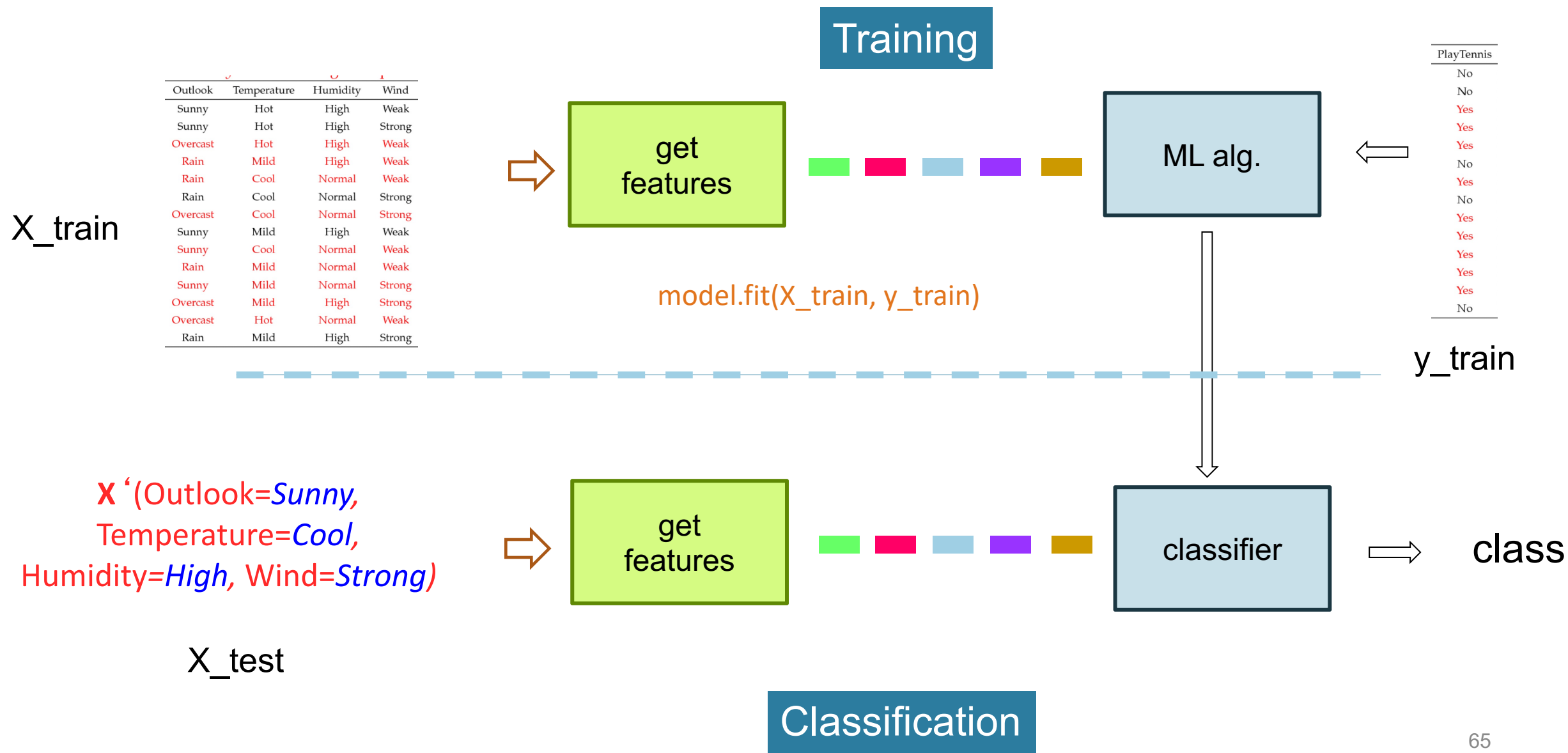


PlayTennis
No
No
Yes
Yes
Yes
No
Yes
No
Yes
Yes
Yes
Yes
Yes
No

Target vector
y_train

model.fit(X_train, y_train)

Two Phases: Training + Classification



HOW A NB CLASSIFIER WORKS

1. *Probability by counting*
2. *Bayes rules*
3. *Naïve Bayes*
4. *Estimation*

[1] Probability by Counting (tan sheet)

	Charges?	Size	Outcome
1	y	small	truthful
2	n	small	truthful
3	n	large	truthful
4	n	large	truthful
5	n	small	truthful
6	n	small	truthful
7	y	small	fraud
8	y	large	fraud
9	n	large	fraud
10	y	large	fraud

Calculate the below values

$$P(\text{truthful}) = ?$$

$$P(\text{large} \mid \text{fraud}) = ?$$

$$P(\text{fraud} \mid y, \text{small}) = ?$$

$$P(\text{fraud} \mid y, \text{large}) = ?$$

$$P(\text{fraud} \mid n, \text{small}) = ?$$

$$P(\text{fraud} \mid n, \text{large}) = ?$$

[1] Probability by Counting

	Charges?	Size	Outcome
1	y	small	truthful
2	n	small	truthful
3	n	large	truthful
4	n	large	truthful
5	n	small	truthful
6	n	small	truthful
7	y	small	fraud
8	y	large	fraud
9	n	large	fraud
10	y	large	fraud

$$P(\text{truthful}) = 6/10$$

$$P(\text{large} \mid \text{fraud}) = 3/4$$

$$P(\text{fraud} \mid y, \text{small}) = 1/2 = 0.5$$

$$P(\text{fraud} \mid y, \text{large}) = 2/2 = 1$$

$$P(\text{fraud} \mid n, \text{small}) = 0/3 = 0$$

$$P(\text{fraud} \mid n, \text{large}) = 1/3 = 0.33$$

[1] Probability by Counting

	Charges?	Size	Outcome
1	y	small	truthful
2	n	small	truthful
3	n	large	truthful
4	n	large	truthful
5	n	small	truthful
6	n	small	truthful
7	y	small	fraud
8	y	large	fraud
9	n	large	fraud
10	y	large	fraud

$$P(\text{ truthful }) = 6/10$$

$$P(\text{ large } | \text{ fraud }) = 3/4$$

$$P(\text{ fraud } | \text{ y, small }) = 1/2$$

$$P(\text{ fraud } | \text{ y, large }) = 2/2$$

$$P(\text{ fraud } | \text{ n, small }) = 0/3$$

$$P(\text{ fraud } | \text{ n, large }) = 1/3$$

[1] Probability by Counting?

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

- Given this limited sample set we can't count, as the chance of an exact match is slim.
- So we look for “partial matches”

Enter Bayes rule ...

The Intuition

$P(\text{yes} \mid \langle \text{Sunny, Cool, High, Strong} \rangle)$

Likelihood

How probable is the evidence
given that our hypothesis is true?

Prior

How probable was our hypothesis
before observing the evidence?

$$P(H | e) = \frac{P(e | H) P(H)}{P(e)}$$

Posterior

How probable is our hypothesis
given the observed evidence?
(Not directly computable)

Marginal

How probable is the new evidence
under all possible hypotheses?
 $P(e) = \sum P(e | H_i) P(H_i)$

Objectives

Motivating Bayes

What is learning?

Review of basic probability

Bayes rule

How a NB classifier works

Bayes in the Real World

BAYES RULE

Bayes' Rule: causal knowledge -> diagnostic knowledge

$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

Causal Knowledge (historic)

- $P(A)$ prior probability of A
- $P(B)$ prior probability of B
- $P(B|A)$ probability of B given that A has occurred
- $P(A|B)$ probability of A given B has occurred

$$P(Measles | Spots) = \frac{P(Spots | Measles) P(Measles)}{P(Spots)}$$

Diagnostic Knowledge

Bayes rule allows us to use historic causal knowledge to make a diagnosis

Derivation of Bayes' rule

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B | A) = \frac{P(B \cap A)}{P(A)}$$

$$P(A | B)P(B) = P(A \cap B) = P(B \cap A) = P(B | A)P(A)$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Aside: Difference between $P(A \cap B)$ and $P(A|B)$

$P(A \cap B)$ = Proportion of things that have
both A and B within all things

$P(\text{BlackHair} \cap \text{BrownEyes}) =$

$P(A|B)$ = Proportion of things that have
A within all things that have B

$P(\text{BlackHair} | \text{BrownEyes}) =$

Aside: Difference between $P(A \cap B)$ and $P(A|B)$

$P(A \cap B)$ = Proportion of things that have
both A and B within all things

$$P(\text{BlackHair} \cap \text{BrownEyes}) = \frac{\text{Number of people that have both}}{\text{Total number of people}}$$

$P(A|B)$ = Proportion of things that have
A within all things that have B

$$P(\text{BlackHair} | \text{BrownEyes}) = \frac{\text{Number of people that have both}}{\text{Number of people with Brown Eyes}}$$

Geometric Intuition behind Bayes' rule

	$\sim A$	A
$\sim B$	w	x
B	y	z

$$P(A) = \frac{x + z}{w + x + y + z}$$

$$P(B) = \frac{y + z}{w + x + y + z}$$

$$P(A|B) = \frac{z}{y + z}$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(B|A) = \frac{z}{x + z} = \frac{\frac{z}{x + z} \cdot \frac{x + z}{w + x + y + z}}{\frac{y + z}{w + x + y + z}} = \frac{z}{y + z}$$

Geometric Intuition behind Bayes' rule

	$\sim A$	A
$\sim B$	w	x
B	y	z

$$P(A) = \frac{x + z}{w + x + y + z}$$

$$P(B) = \frac{y + z}{w + x + y + z}$$

$$P(A|B) = \frac{z}{y + z}$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$\begin{aligned}
 P(B|A) &= \frac{z}{x + z} \\
 &= \frac{\frac{z}{x + z} \cdot \frac{x + z}{w + x + y + z}}{\frac{y + z}{w + x + y + z}} \\
 &= \frac{z}{y + z}
 \end{aligned}$$

Back to ... Two fair coins and one double headed



You randomly pick one coin, toss it 3 times and get **3 heads**.

What is the probability now that you had picked the double headed coin?

$$P(DHcoin | 3heads) = ?$$

Back to ... Two fair coins and one double headed



You randomly pick one coin, toss it 3 times and get **3 heads**.

What is the probability now that you had picked the double headed coin?

$$P(DHcoin | 3heads) = \frac{P(3heads|DHcoin)P(DHcoin)}{P(3heads)}$$

Back to ... Two fair coins and one double headed



You randomly pick one coin, toss it 3 times and get **3 heads**.

What is the probability now that you had picked the double headed coin?

$$P(DHcoin | 3heads) = \frac{P(3heads|DHcoin)P(DHcoin)}{P(3heads)}$$

$$= \frac{1 * \frac{1}{3}}{\frac{1}{3}(\frac{1}{2} * \frac{1}{2} * \frac{1}{2}) + \frac{1}{3}(\frac{1}{2} * \frac{1}{2} * \frac{1}{2}) + \frac{1}{3}(1 * 1 * 1)} = \frac{1}{\frac{1}{8} + \frac{1}{8} + 1} = \frac{8}{10} = \frac{4}{5}$$

2. Exact Bayes

	Charges?	Size	Outcome
1	y	small	truthful
2	n	small	truthful
3	n	large	truthful
4	n	large	truthful
5	n	small	truthful
6	n	small	truthful
7	y	small	fraud
8	y	large	fraud
9	n	large	fraud
10	y	large	fraud

Complete
bottom of
front page of
tan handout

$$P(\text{fraud} \mid n, \text{large}) = \frac{P(n, \text{large} \mid \text{fraud}) * P(\text{fraud})}{P(n, \text{large})}$$

2. Exact Bayes

	Charges?	Size	Outcome
1	y	small	truthful
2	n	small	truthful
3	n	large	truthful
4	n	large	truthful
5	n	small	truthful
6	n	small	truthful
7	y	small	fraud
8	y	large	fraud
9	n	large	fraud
10	y	large	fraud

$$P(\text{fraud} \mid n, \text{large}) = \frac{P(n, \text{large} \mid \text{fraud}) * P(\text{fraud})}{P(n, \text{large})} = 1/4$$

2. Exact Bayes

	Charges?	Size	Outcome
1	y	small	truthful
2	n	small	truthful
3	n	large	truthful
4	n	large	truthful
5	n	small	truthful
6	n	small	truthful
7	y	small	fraud
8	y	large	fraud
9	n	large	fraud
10	y	large	fraud

$$P(\text{fraud} \mid n, \text{large}) = \frac{P(n, \text{large} \mid \text{fraud}) * P(\text{fraud})}{P(n, \text{large})} = \frac{1/4 * 4/10}{}$$

2. Exact Bayes

	Charges?	Size	Outcome
1	y	small	truthful
2	n	small	truthful
3	n	large	truthful
4	n	large	truthful
5	n	small	truthful
6	n	small	truthful
7	y	small	fraud
8	y	large	fraud
9	n	large	fraud
10	y	large	fraud

$$P(\text{fraud} \mid n, \text{large}) = \frac{P(n, \text{large} \mid \text{fraud}) * P(\text{fraud})}{P(n, \text{large})} = \frac{1/4 * 4/10}{3/10}$$

2. Exact Bayes

	Charges?	Size	Outcome
1	y	small	truthful
2	n	small	truthful
3	n	large	truthful
4	n	large	truthful
5	n	small	truthful
6	n	small	truthful
7	y	small	fraud
8	y	large	fraud
9	n	large	fraud
10	y	large	fraud

$$P(\text{fraud} \mid n, \text{large}) = \frac{P(n, \text{large} \mid \text{fraud}) * P(\text{fraud})}{P(n, \text{large})} = \frac{1/4 * 4/10}{3/10} = \mathbf{0.33}$$

2. Exact Bayes?

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Stuck again !!!

$$\begin{aligned}P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) \\&= P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{yes}) * P(\text{yes}) / \\&\quad P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)\end{aligned}$$

See back of handout

NAÏVE BAYES

3. Naïve Bayes

$$\begin{aligned} P(\text{fraud} \mid n, \text{large}) &= \frac{P(n, \text{large} \mid \text{fraud}) * P(\text{fraud})}{P(n, \text{large})} \\ &= \frac{P(n \mid \text{fraud}) * P(\text{large} \mid \text{fraud}) * P(\text{fraud})}{P(n, \text{large})} \\ &= \frac{P(n \mid \text{fraud}) * P(\text{large} \mid \text{fraud}) * P(\text{fraud})}{P(n, \text{large} \mid \text{fraud}) * P(\text{fraud}) + P(n, \text{large} \mid \text{truthful}) * P(\text{truthful})} \\ &= \frac{P(n \mid \text{fraud}) * P(\text{large} \mid \text{fraud}) * P(\text{fraud})}{P(n \mid \text{fraud}) * P(\text{large} \mid \text{fraud}) * P(\text{fraud}) + P(n \mid \text{truthful}) * P(\text{large} \mid \text{truthful}) * P(\text{truthful})} \end{aligned}$$

$$P(\text{fraud} \mid n, \text{large}) = \frac{(1/4 * 3/4 * 4/10)}{(1/4 * 3/4 * 4/10) + (5/6 * 2/6 * 6/10)} = \mathbf{0.31}$$

3. Naïve Bayes! Finally ...

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)$

$$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) = \frac{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{yes}) * P(\text{yes})}{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)}$$

The two probabilities we are interested in

$$P(\text{yes} \mid \langle \text{Sunny, Cool, High, Strong} \rangle) = \frac{P(\langle \text{Sunny, Cool, High, Strong} \rangle \mid \text{yes}) * P(\text{yes})}{P(\langle \text{Sunny, Cool, High, Strong} \rangle)}$$

$$P(\text{no} \mid \langle \text{Sunny, Cool, High, Strong} \rangle) = \frac{P(\langle \text{Sunny, Cool, High, Strong} \rangle \mid \text{no}) * P(\text{no})}{P(\langle \text{Sunny, Cool, High, Strong} \rangle)}$$

But we are really interested only in comparing them

$$P(\text{yes} \mid \langle \text{Sunny, Cool, High, Strong} \rangle) \approx \frac{P(\langle \text{Sunny, Cool, High, Strong} \rangle \mid \text{yes}) * P(\text{yes})}{P(\langle \text{Sunny, Cool, High, Strong} \rangle)}$$

$$P(\text{no} \mid \langle \text{Sunny, Cool, High, Strong} \rangle) \approx \frac{P(\langle \text{Sunny, Cool, High, Strong} \rangle \mid \text{no}) * P(\text{no})}{P(\langle \text{Sunny, Cool, High, Strong} \rangle)}$$

Hence we don't need to compute the denominators!

Calculating and Estimating Probabilities

1. Counting

$$\frac{\text{\# of Outcomes in event E}}{\text{\# of Outcomes in Sample Space}}$$

2. Exact Bayes' rule

$$P(C | f_1, f_2) = \frac{P(f_1, f_2 | C) * P(C)}{P(f_1, f_2)}$$

3. Naïve (approximate) Bayes' rule

- using the independence assumption

$$P(C | f_1, f_2) = \frac{P(f_1 | C) * P(f_2 | C) * P(C)}{P(f_1, f_2)}$$

-

$$P(C_1 | f_1 f_2) = \frac{P(f_1 | C_1) * P(f_2 | C_1) * P(C_1)}{P(f_1 | C_1) * P(f_2 | C_1) * P(C_1) + P(f_1 | C_2) * P(f_2 | C_2) * P(C_2)}$$

4. Relative Estimation

$$P(f_1 | C_1) * P(f_2 | C_1) * P(C_1)$$

Green handout

Objectives

Motivating Bayes

What is learning?

Review of basic probability

Bayes rule

How a NB classifier works

Bayes in the Real World

HOW A NB CLASSIFIER WORKS

Back to our example ...

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)$

$$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) = \frac{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{yes}) * P(\text{yes})}{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)}$$

Focusing on the numerators ... we get ...

$$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) \approx \frac{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{yes}) * P(\text{yes})}{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)}$$

$$P(\text{no} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) \approx \frac{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{no}) * P(\text{no})}{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)}$$

Exercise: Compute and Compare (Green handout)

$$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) \approx P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{yes}) * P(\text{yes})$$

$$P(\text{Yes} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{Yes}) P(\text{Cool} \mid \text{Yes}) P(\text{High} \mid \text{Yes}) P(\text{Strong} \mid \text{Yes})] * P(\text{Yes})$$

$$P(\text{no} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) \approx P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{no}) * P(\text{no})$$

$$P(\text{No} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{No}) P(\text{Cool} \mid \text{No}) P(\text{High} \mid \text{No}) P(\text{Strong} \mid \text{No})] * P(\text{No})$$

1. Compute the tables for each feature (Outlook, Temperature, Humidity, Wind)
2. Compute the probability of each category (yes, no)
3. Compute the individual probabilities

Complete these tables (Green handout)

Outlook	Play=Yes	Play=No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rain	3/9	2/5

Temperature	Play=Yes	Play=No
Hot		
Mild		
Cool		

Humidity	Play=Yes	Play=No
High		
Normal		

Wind	Play=Yes	Play=No
Strong		
Weak		

$$P(\text{Play=Yes}) = 9/14$$

$$P(\text{Play=No}) = 5/14$$

From these tables

Outlook	Play=Yes	Play=No
<i>Sunny</i>	2/9	3/5
<i>Overcast</i>	4/9	0/5
<i>Rain</i>	3/9	2/5

Temperature	Play=Yes	Play=No
<i>Hot</i>	2/9	2/5
<i>Mild</i>	4/9	2/5
<i>Cool</i>	3/9	1/5

Humidity	Play=Yes	Play=No
<i>High</i>	3/9	4/5
<i>Normal</i>	6/9	1/5

Wind	Play=Yes	Play=No
<i>Strong</i>	3/9	3/5
<i>Weak</i>	6/9	2/5

$$P(\text{Play=Yes}) = 9/14$$

$$P(\text{Play=No}) = 5/14$$

Relative Estimation

Outlook	Play=Yes	Play=No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rain	3/9	2/5

Temperature	Play=Yes	Play=No
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	Play=Yes	Play=No
High	3/9	4/5
Normal	6/9	1/5

Wind	Play=Yes	Play=No
Strong	3/9	3/5
Weak	6/9	2/5

$x' = (\text{Outlook}=\textit{Sunny}, \text{Temperature}=\textit{Cool}, \text{Humidity}=\textit{High}, \text{Wind}=\textit{Strong})$

Look up from tables

$P(\text{Outlook}=\textit{Sunny} \mid \text{Play}=\textit{Yes}) = 2/9$

$P(\text{Temperature}=\textit{Cool} \mid \text{Play}=\textit{Yes}) = 3/9$

$P(\text{Humidity}=\textit{High} \mid \text{Play}=\textit{Yes}) = 3/9$

$P(\text{Wind}=\textit{Strong} \mid \text{Play}=\textit{Yes}) = 3/9$

$P(\text{Play}=\textit{Yes}) = 9/14$

$P(\text{Outlook}=\textit{Sunny} \mid \text{Play}=\textit{No}) = 3/5$

$P(\text{Temperature}=\textit{Cool} \mid \text{Play}=\textit{No}) = 1/5$

$P(\text{Humidity}=\textit{High} \mid \text{Play}=\textit{No}) = 4/5$

$P(\text{Wind}=\textit{Strong} \mid \text{Play}=\textit{No}) = 3/5$

$P(\text{Play}=\textit{No}) = 5/14$

$P(\text{Play}=\textit{Yes}) = 9/14 \quad P(\text{Play}=\textit{No}) = 5/14$

Relative Estimation

Outlook	Play=Yes	Play=No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rain	3/9	2/5

Temperature	Play=Yes	Play=No
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	Play=Yes	Play=No
High	3/9	4/5
Normal	6/9	1/5

Wind	Play=Yes	Play=No
Strong	3/9	3/5
Weak	6/9	2/5

$\mathbf{x}' = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$

Look up from tables

$$P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{Yes}) = 2/9$$

$$P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Humidity}=\text{High} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Play}=\text{Yes}) = 9/14$$

$$P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{No}) = 3/5$$

$$P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{No}) = 1/5$$

$$P(\text{Humidity}=\text{High} \mid \text{Play}=\text{No}) = 4/5$$

$$P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{No}) = 3/5$$

$$P(\text{Play}=\text{No}) = 5/14$$

$$P(\text{Yes} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{Yes}) P(\text{Cool} \mid \text{Yes}) P(\text{High} \mid \text{Yes}) P(\text{Strong} \mid \text{Yes})] \cdot$$

$$P(\text{Play}=\text{Yes})$$

$$[2/9 \cdot 3/9 \cdot 3/9 \cdot 3/9] \cdot 9/14 = 0.0053$$

$$P(\text{No} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{No}) P(\text{Cool} \mid \text{No}) P(\text{High} \mid \text{No}) P(\text{Strong} \mid \text{No})] \cdot P(\text{Play}=\text{No})$$

$$3/5 \cdot 1/5 \cdot 4/5 \cdot 3/5 \cdot 5/14 = 0.0206$$

$$P(\text{Play}=\text{Yes}) = 9/14 \quad P(\text{Play}=\text{No}) = 5/14$$

Given $P(\text{Yes} \mid \mathbf{x}') < P(\text{No} \mid \mathbf{x}')$, we classify $\mathbf{x}'$ to be “No”

Objectives

Motivating Bayes

What is learning?

Review of basic probability

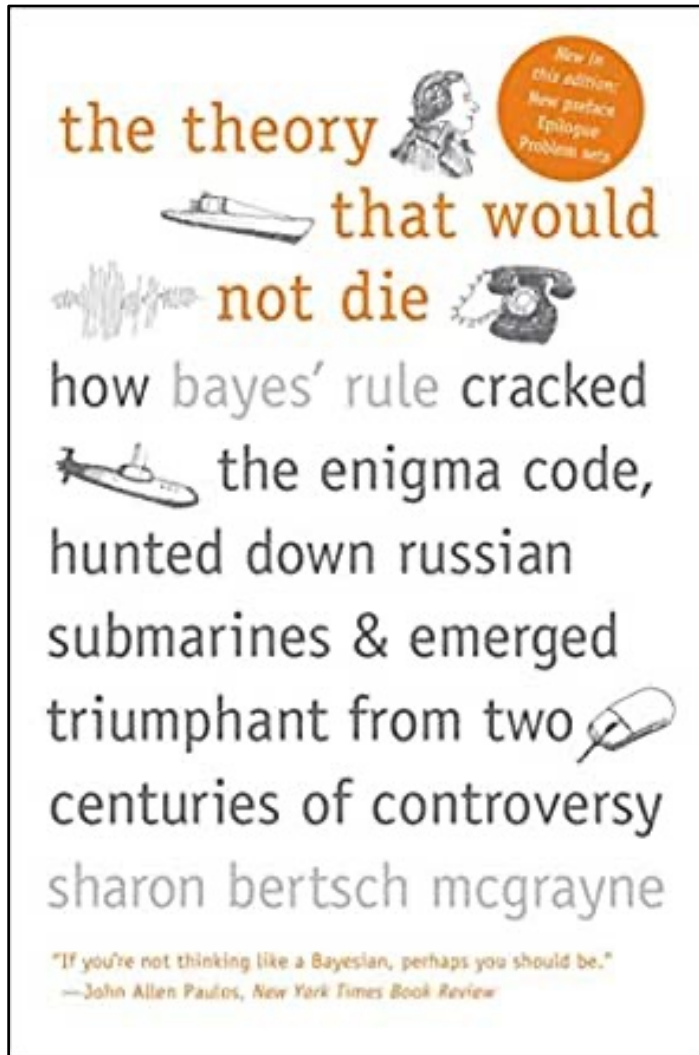
Bayes rule

How a NB classifier works

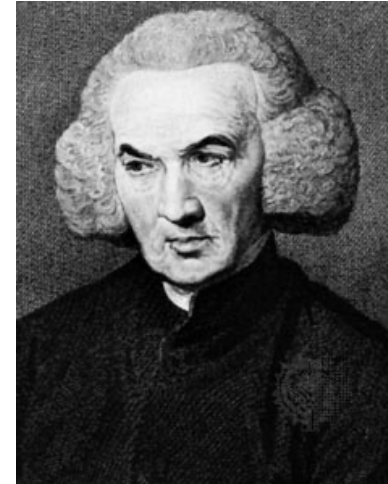
Bayes in the Real World

BAYES IN THE REAL WORLD

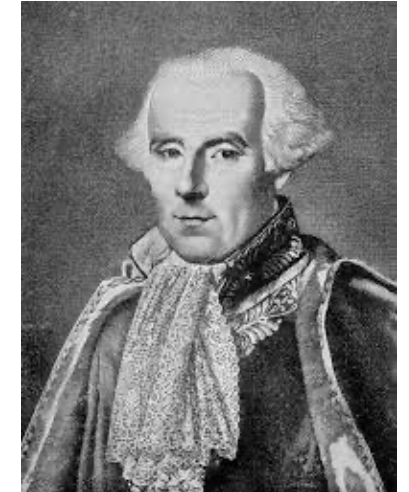
The Tumultuous and Fascinating History of Bayes



Thomas Bayes



Richard Price



Pierre-Simon Laplace



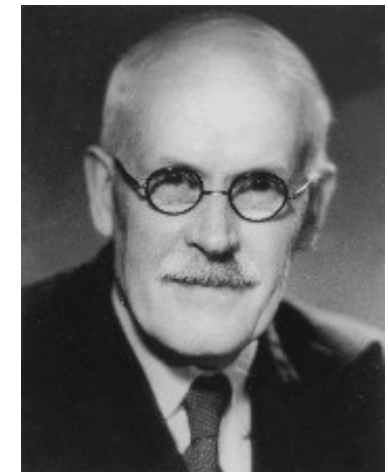
Ronald Fisher



Karl Pearson



Jerzy Neyman

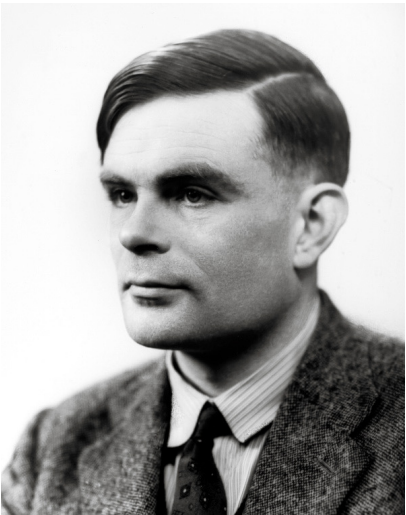


Harold Jeffreys

Bayes in the Real World (to name a few)



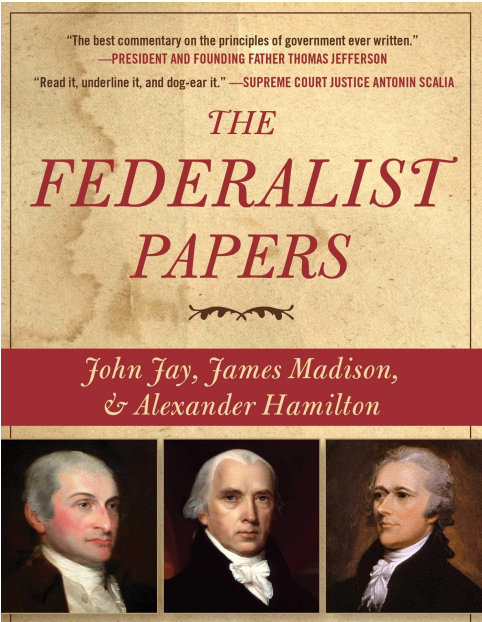
Enigma Machine



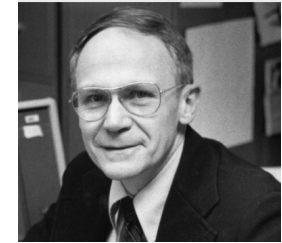
Alan Turing



Lost submarine USS Scorpion



Authorship resolution



Mosteller & Wallace

How Statisticians Found Air France Flight 447 Two Years After It Crashed Into Atlantic

After more than a year of unsuccessful searching, authorities called in an elite group of statisticians. Working on their recommendations, the next search found the wreckage just a week later.

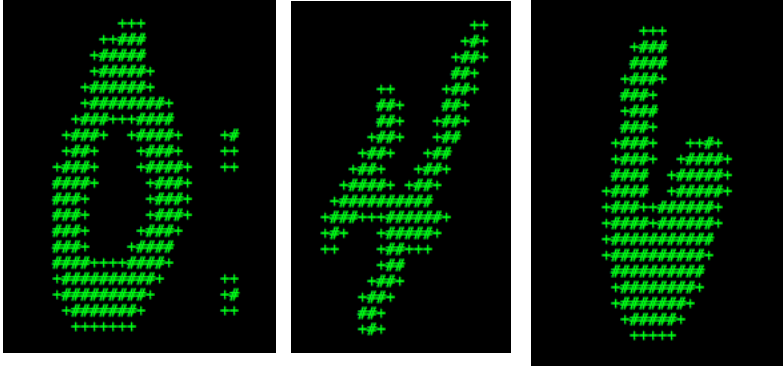


Lost B28 Hydrogen bomb



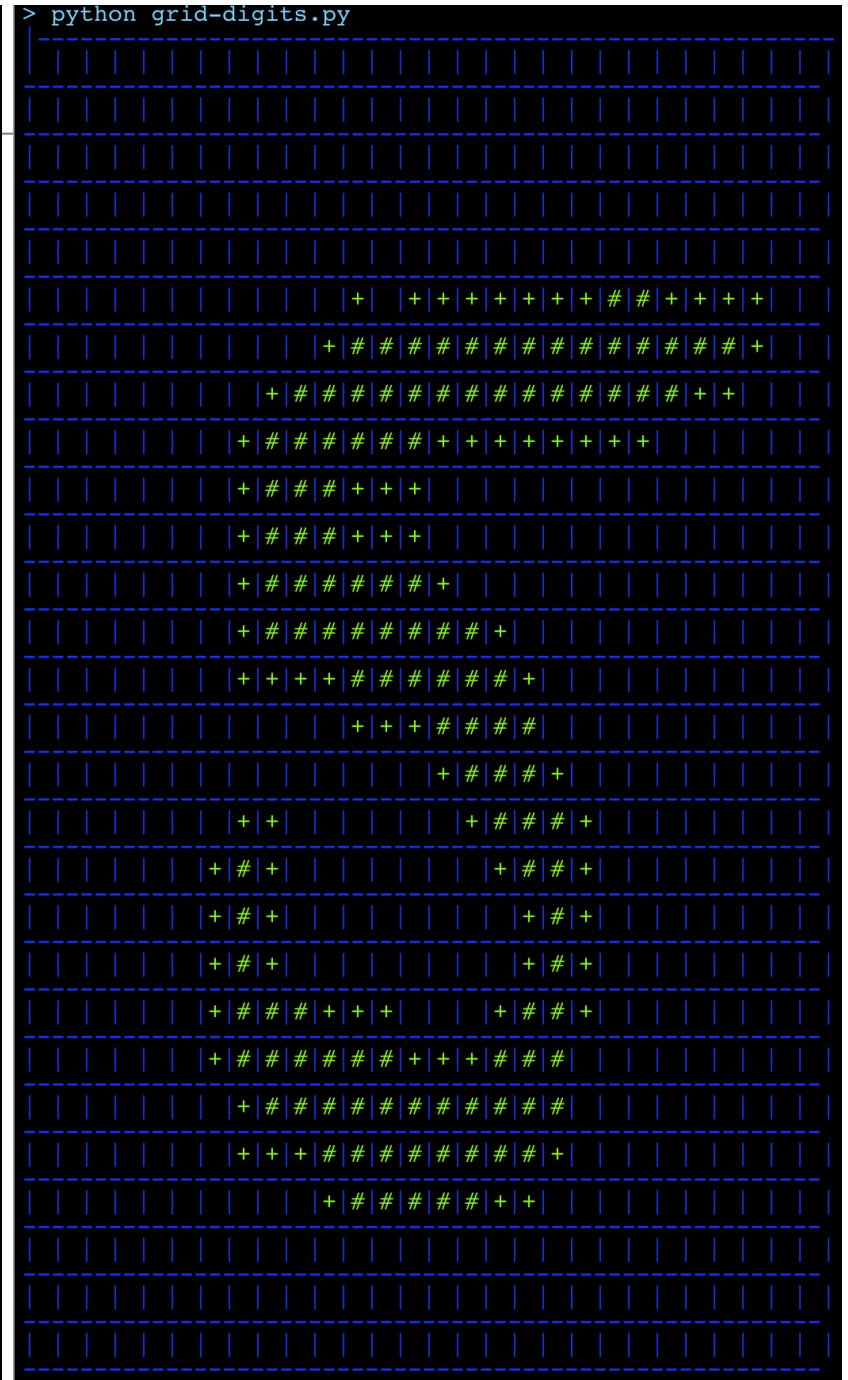
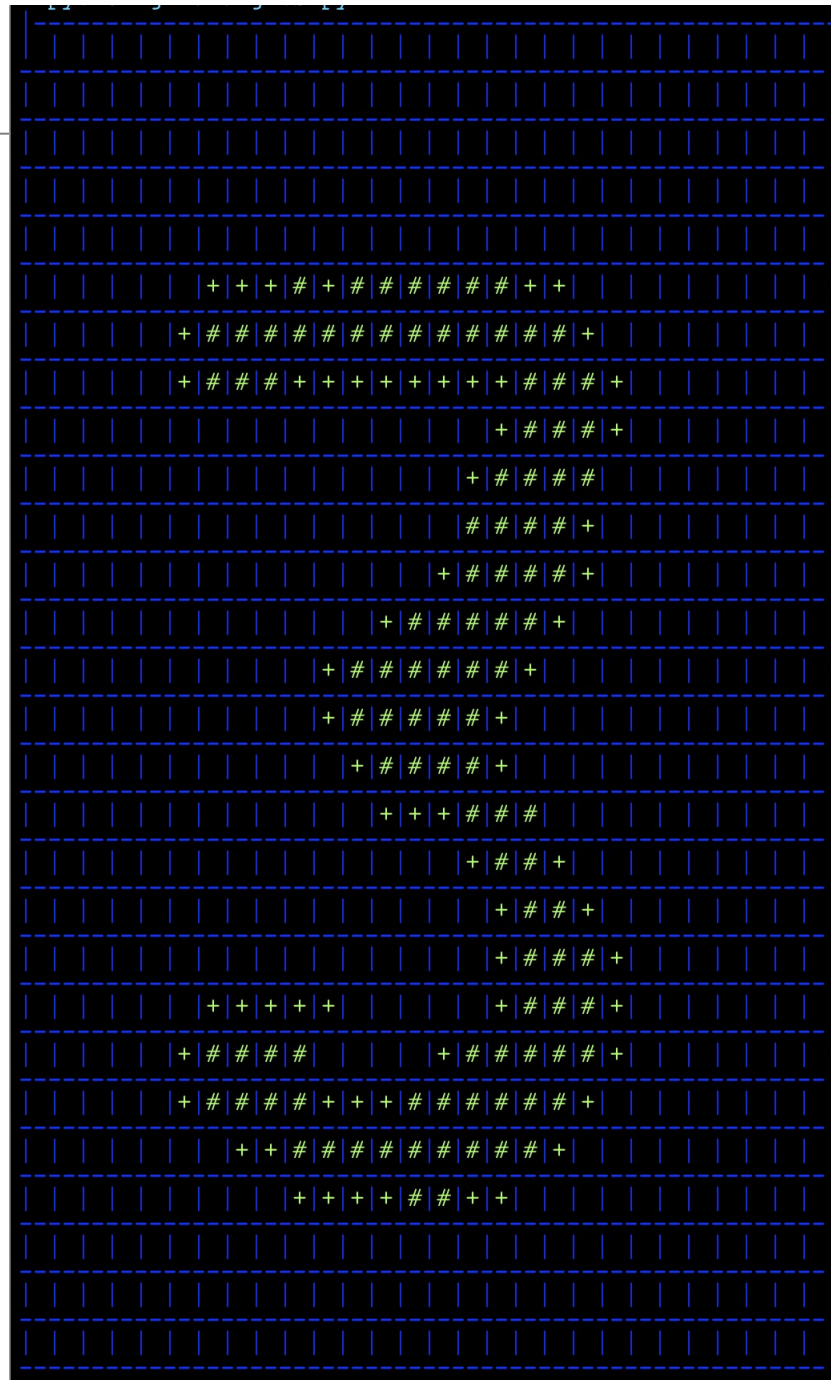
DEMO

MNIST Database

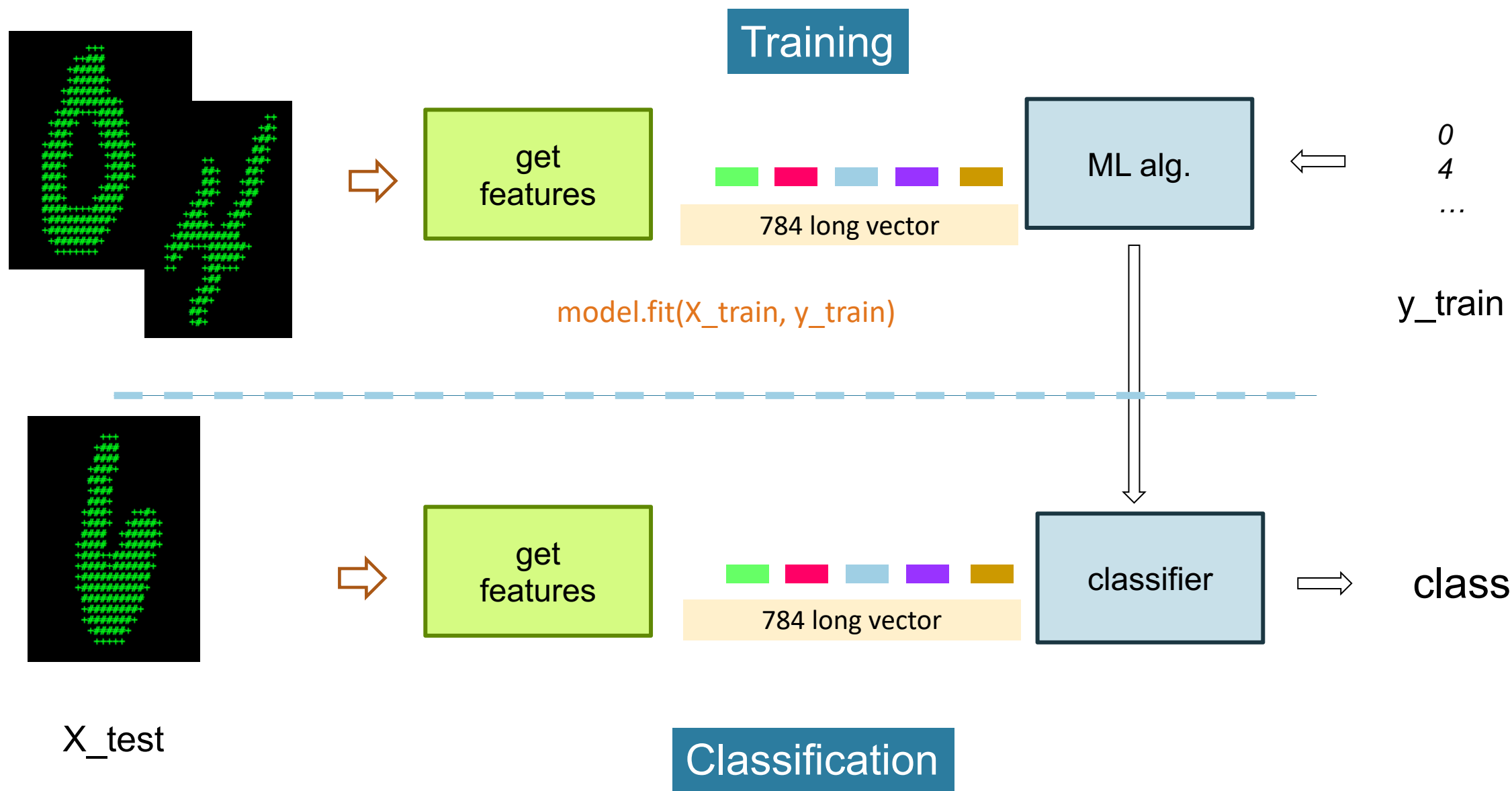


Training: 5000 images
Testing: 1000 images

*Each "image" is a
28x28 ascii grid
784 features*



Two Phases: Training + Classification



What we want ...

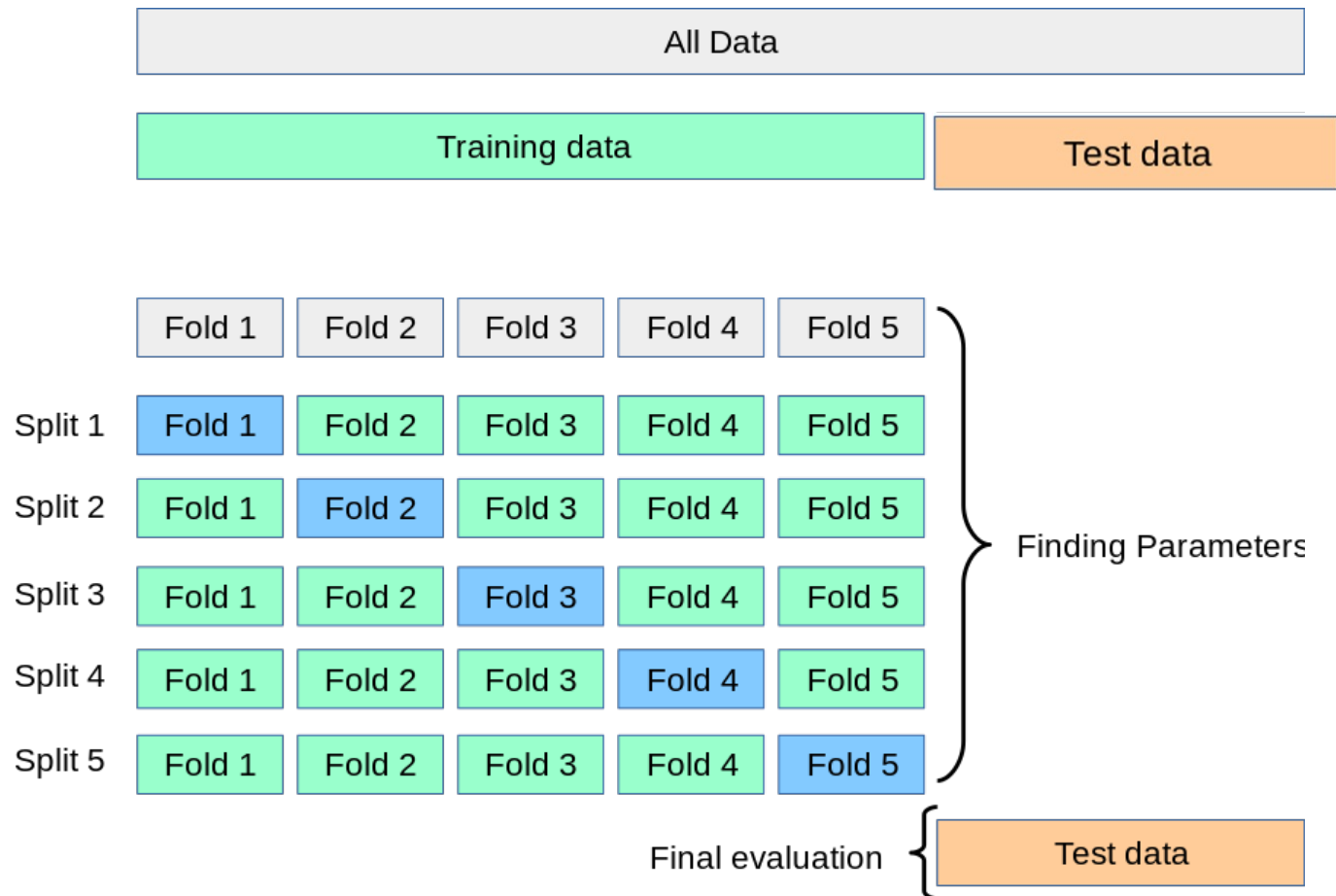
- $P(X \mid f_1, f_2, f_3, \dots, f_{784})$ for $X = 0, 1, 2, \dots, 9$

- What we did Friday

$P(X \mid f_1, f_2, f_3, \dots, f_{784}, f_{785})$ for $X = 0, 1, 2, \dots, 9$

Train-Test split vs. k-fold cross-validation

```
X_train, X_test,  
y_train, y_test =  
    train_test_split(X, y,  
                    test_size=0.2,  
                    random_state=1)
```



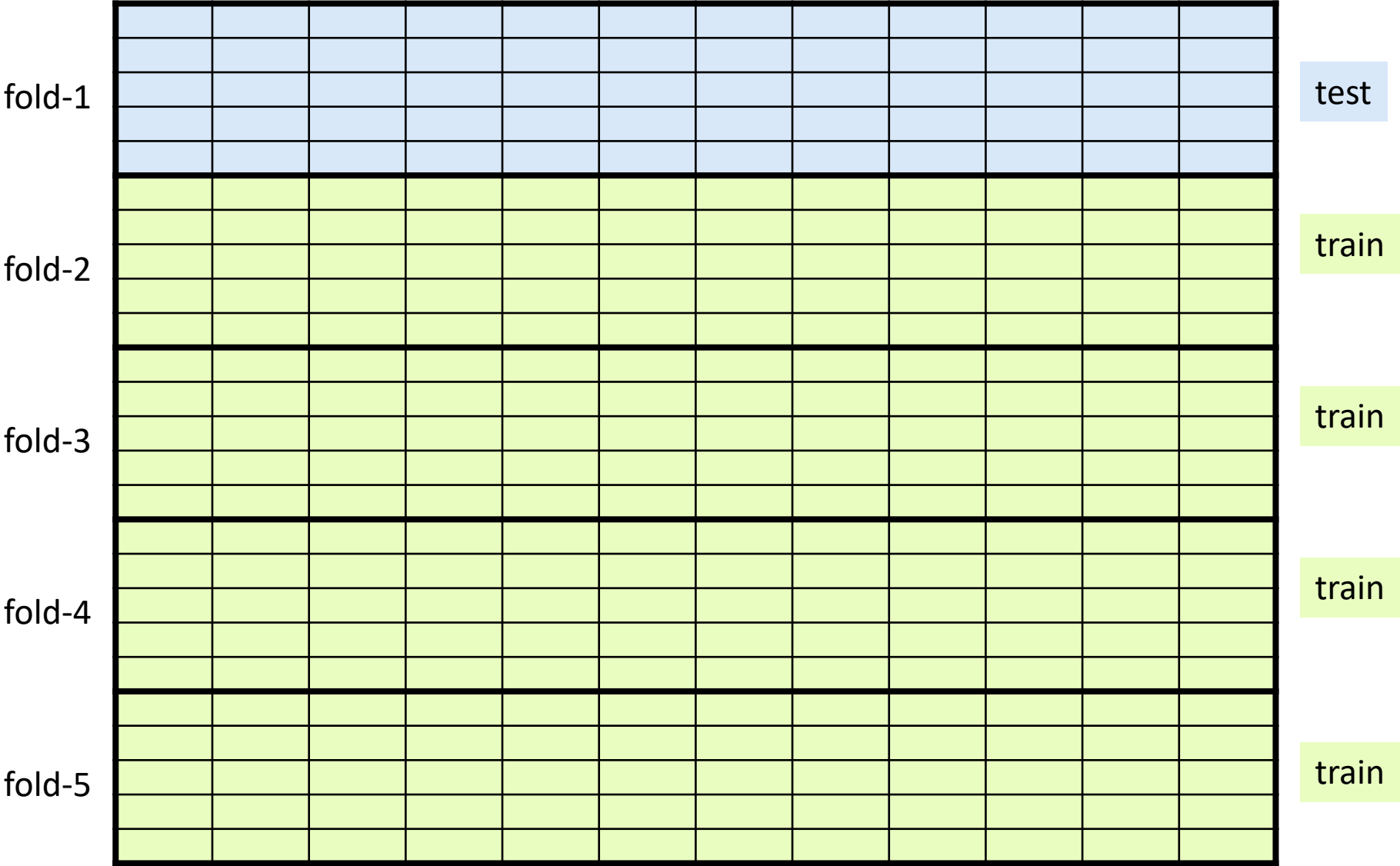
https://en.wikipedia.org/wiki/Training,_validation,_and_test_data_sets

`sklearn.model_selection.RepeatedStratifiedKFold`

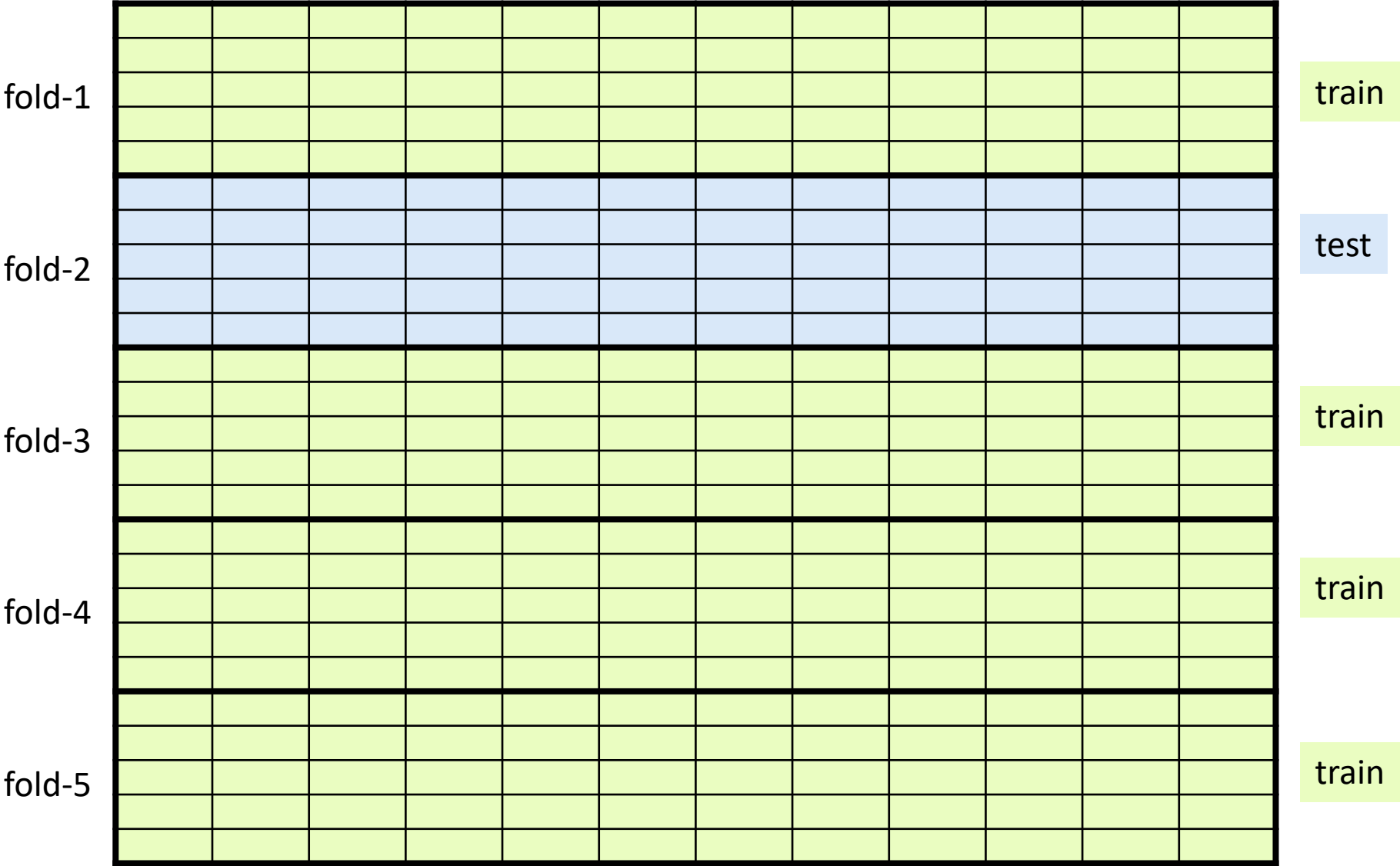
https://scikit-learn.org/stable/modules/cross_validation.html

[illegible]

k-fold cross-validation



k-fold cross-validation



Temporal Cross Validation

